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Parts I & II
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AFSC 3E1X1
CIVIL ENGINEER HVAC/R
CFETP

BASIC MASTER SENIOR

CAREER FIELD EDUCATION AND TRAINING PLAN





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PART I

Preface

This Career Field Education and Training Plan (CFETP) is a comprehensive education and training document that identifies life-cycle education/training requirements and training support resources for the Engineering specialty. The CFETP will provide our personnel with a clear career path to success and instill rigor in all aspects of our career field training.

The CFETP consists of two parts used by the supervisor to plan, manage, and control training within the career field.

Part I provides information necessary for overall management of the specialty.

- Section A provides general information on how to use the CFETP.
- Section B identifies information on career field progression, duties and responsibilities, training strategies, and the career field path.
- Section C associates each skill level with specialty qualifications (knowledge, education, and training).
- Section D indicates resource constraints.
- Section E identifies transition/training guidance requirements for SSgt through MSgt.

Part II includes the following:

- Section A identifies the Specialty Training Standard (STS) to include duties, tasks, and technical references to support Air Education and Training Command (AETC) conducted training, contingency courses, and correspondence course requirements.
- Section B contains the course objective list and training standards supervisors will use to determine if an Airman has satisfied training requirements.
- Section C identifies available support materials.
- Section D identifies a training course index that supervisors can use to determine resources available to support training. Included here are the exportable courseware, mandatory, and supplemental courses.
- Section E identifies MAJCOM-unique training requirements supervisors can use to determine additional training required for the associated qualification needs.
- Section F identifies home station training references and course materials required for this specialty to support contingency/wartime training.

Using guidance provided in the CFETP will ensure individuals in this specialty receive effective and efficient training at the appropriate point in their careers. This plan will enable us to train today's workforce for tomorrow's jobs. At the unit level, supervisors and trainers will use Part II to identify, plan, and conduct training commensurate with the overall goals of this guide.

Abbreviations and Terms Explained

Air Force Civil Engineer Center (AFCEC). The focal point for all Civil Engineer (CE) training development. Assigned to AFCEC are all the AFSC Readiness and Training Managers (RTMs).

Air Force Career Field Manager (AFCFM). An individual on the Air Staff charged with the responsibility for overseeing all training and career field management aspects of an Air Force specialty or group of specialties.

Department of the Air Force Enlisted Classification Directory (DAFECD). The official directory for all military enlisted classification descriptions, codes, and identifiers. The specialty descriptions and codes used to identify each Air Force job; the DAFECD describes the minimum mandatory qualifications for personnel to fill these jobs. The updated DAFECD is available at the AF Personnel Center's website located at **myFSS**.

Air Force Credentialing Opportunities Online (AF COOL) Program. Access the AF COOL Program through the **Air Force Virtual Education Center (AFVEC)** website. The site provides a research tool designed to increase an Airman's awareness of national professional credentialing and CCAF education opportunities available for all Air Force occupational specialties.

Air Force Institute of Technology (AFIT). Provides vital, relevant, and connected education that enables Airmen to be ready engineers and great leaders who know how to build sustainable installations to last while leading the change for the Civil Engineer career field. Access the AFIT course list in the **AFIT Civil Engineer School Course Catalog**.

Air Force Job Qualification Standard/Command Job Qualification Standard (AFJQS/CJQS). An AFJQS is a comprehensive task list that describes a particular job type or duty position; supervisors use it to document task qualifications. The tasks on the AFJQS/CJQS are common to all personnel serving in the described duty position.

Air Force Qualification Training Package (AFQTP). A required instructional package designed for use at the unit to qualify, or aid qualification, in a duty position, program, or on a piece of equipment. AFQTPs identify the Air Force's standardized method for performing the task. The AFQTP are paper-based, computer-based, in other audiovisual media formats, or all three. Each Airman must use AFQTPs to satisfy a particular training requirement. AFQTPs for the 3E1X1 AFSC are located on **myLearning** and on **CE DASH**.

Career Field Education and Training Plan (CFETP). A comprehensive, multipurpose document encapsulating the entire spectrum of education and training for a career field. It outlines a logical growth plan that includes training resources designed to make career field training identifiable, eliminate duplication, and ensure this training is budget defensible.

Chief, Readiness and Training Division. This individual at the AFCEC and responsible for readiness and training within the 3E0 to 3E6 AFSCs.

Commercial Off The Shelf (COTS). Commercially procured training products.

Competencies. The combination of Knowledge, Skills, Abilities, and Other Characteristics (KSAOs) that manifest in an observable, measurable pattern of **behaviors**. **Note:** competencies will provide Airmen with a clear sight picture of the education, training and experiences they will need to perform successfully on the job.

Continuum of Learning (CoL). Career-long process of individual development where challenging experiences are combined with education and training to produce Airmen who possess the tactical expertise, operational competence, strategic vision, and joint proficiency to lead and execute the full spectrum of Air Force and joint missions. Designed to integrate developmental opportunities through a common taxonomy to produce adaptable and knowledgeable enabled Airmen for today and tomorrow.

Core Tasks (5^ or 7^). Mandatory tasks, which the AFCFM has identified as the minimum qualification requirements within an Air Force specialty or duty position. These tasks exemplify the essence of the career field.

Critical Tasks. Tasks the work center supervisor identifies as detrimental to mission accomplishment if not performed correctly. Critical tasks may or may not be the same as core tasks but are mandatory if identified as "critical" to the position by the supervisor or work center.

Diamond Tasks (♦). Diamond tasks are extremely important to the career field. Diamond tasks are the same as core tasks with one exception: equipment shortfalls at most locations have created problems with the actual hands-on training/certification of these tasks. In instances where required equipment is not available for instruction, completing the AFQTP is required for upgrade and qualification training. Airmen must accomplish hands-on certification at the first opportunity when equipment is available.

Distance Learning (DL). Includes Video Tele-seminar (VTS), Video Tele-training (VTT), and WBT. Formal courses that a training wing or a contractor develops for export to a field location (in place of resident training) for trainees to complete without the on-site support of a formal school instructor. For instance, the Air Force Institute of Technology, Air University, and Air Education Training Command offer online courses.

Duty Position Tasks. The tasks assigned to an individual for the position currently held. These include as a minimum all core tasks, critical tasks and any other tasks assigned by the supervisor.

Enlisted Professional Military Education (EPME). Introduces Airmen to appropriate institutional competencies at specific milestones throughout their career and includes two phases: Basic and Resident. Basic EPME requirements are via distance learning courses to establish a foundation for continued development. Resident attendance is not duplicative of the basic EPME requirements but builds upon the competencies obtained to achieve higher proficiency levels.

Enlisted Promotions Requirements Course Catalog (EPRRC). The EPRRC lists all enlisted promotion tests authorized for administration and the study references associated with these tests. Every question on a promotion test comes from one of the publications listed in the catalog. The site only lists publications that support questions on a given promotion test. The catalog also contains administrative and special instructions for test control officers. The EPRRC is available at the **Airman Promotion Tools Testing** website.

Education and Training Course Announcements (ETCA). Contains specific MAJCOM procedures, fund cite instructions, reporting instructions, and listings for formal courses conducted or managed by MAJCOMs or Field Operating Agencies (FOAs).

Expeditionary Combat Support-Training Certification Center (ECS-TCC). The Air Force Reserve Command manages this training center.

Foundational Competencies. Used to implement the continuum of learning. These leadership competencies are what the AF expects of all Airmen. These competencies allow Airmen to operate successfully in a constantly changing environment in which we function.

Interservice Training Review Organization (ITRO). Apprentice or advanced training that is provided in a Joint Service setting.

Initial Skills Training. Air Force Specialty (AFS). Specific training an individual receives upon entry into the Air Force or upon retraining into this specialty for the award of the 3-skill level. For our career field, the 366 TRS at Sheppard AFB, TX provides the training.

MAJCOM Functional Managers (MFMs). Senior leaders designated by the appropriate functional authority provide day-to-day management and responsibility over specific functional communities at the MAJCOM, Field Operating Agency (FOA), Direct Reporting Unit (DRU), or Air Reserve Component (ARC) level. While they should maintain an institutional focus regarding resource deployment and distribution, MFMs are responsible for ensuring their specialties are equipped, developed, and sustained to meet the future needs of the total Air Force mission.

myLearning. Anytime, anywhere, learning for the Civil Engineer Community, consisting of instructional and skill-level awarding course material specific to an AFSC.

myTraining. The Total Force Training Record for Airmen that provides the capability through 21st-century learning tools to manage the training lifecycle for Total Force personnel.

Occupational Analysis Report (OAR). A detailed report showing the results of an occupational survey of tasks performed within a particular AFS. The CFM uses the information collected from this survey to make changes to upgrade training and Weighted Airman Promotion Exams.

On-the-Job Training (OJT). Hands-on, over-the-shoulder training conducted to certify personnel in both upgrade (skill-level award) and job qualification (duty position certification) training.

Occupational Competencies. A set of competencies required of all Airmen within a specific workforce category (a group of functions requiring similar work, i.e., troubleshooting). They describe technical/functional skills, knowledge, abilities, behaviors, and other characteristics needed to perform that function's mission successfully.

Proficiency Training. Additional training, either in-residence or exportable advanced training courses, or on-the-job training, provided to personnel to increase their skills and knowledge beyond the minimum training required for upgrade.

Qualification Training (QT). Hands-on task performance training designed to qualify an Airman in a specific duty position. This training occurs during and after upgrade training. Qualification training provides the performance and knowledge training required to do the job.

Readiness and Training Manager (RTM). An individual assigned to the Air Force Civil Engineer Center (AFCEC) who serves as the HAF/A4C execution agent for all training and some career field management aspects of a specific Air Force Civil Engineer specialty.

Regional Training Site (RTS). The Air National Guard manages these Total Force training centers.

Resource Constraints. Resource deficiencies, such as money, facilities, time, manpower, or equipment that precludes desired delivery of training.

Specialty Training Requirements Team (STRT). Before a U&TW, the AFCFM and a team of Subject Matter Experts (SMEs) from each MAJCOM meet to determine education and training requirements (formal and on-the-job training) for an Air Force Specialty. They use the STRT to create or revise training standards for all types of training. The team finalizes the STS, DAFECD specialty description, and supplemental Course Training Standards (CTSs).

Specialty Training Standard (STS). Describes skills and knowledge that Airmen in a particular AFS need on the job. It further serves as a contract between AETC and the user to show the overall training course requirements for an AFS taught in resident or non-resident.

Subject Matter Expert (SME). An individual with expertise in a particular subject matter tasked to represent the subject matter to an individual or group for technical accuracy.

Supplemental Training. A formal course that provides individuals qualified in one or more positions of their Air Force Specialty (AFS) with additional skills/knowledge to enhance their expertise in the career field. Training is for selected career airmen at the advanced level of the AFS.

Task Qualification Training (TQT). Training conducted after Airmen have received Chemical, Biological, Radiological, and Nuclear Defense classroom training. Individuals perform wartime mission essential tasks in a simulated wartime environment while wearing full ground crew individual protective equipment or aircrew individual protective equipment.

Total Force. Organizations, units, and individuals that comprise the Air Forces' resources for meeting its mission. This includes civilian personnel, Active Duty, Air National Guard, and Air Force Reserve Command military personnel.

Training Planning Team (TPT). Comprised of the same personnel as a U&TW, TPTs are more involved in training development and the range of issues, not found in normal U&TWs.

Upgrade Training (UGT). The CFETP identifies the mandatory courses, task qualification requirements, and correspondence course completion requirements for awarding the 5-, 7-, and 9-skill levels.

Utilization and Training Workshop (U&TW). An executive decision meeting to vote on funding (Course Resource Estimates) for instructor authorizations, equipment, and facilities needed to support any new or revised training coming from the STRT. They will also determine which organizations will furnish resources, establish commitment and delivery dates in writing, document equipment availability dates, identify any problems and establish training delivery dates.

Vectored Positions. Key SNCO positions in your career field. To learn more about vectored positions, go to **MyVECTOR**.

Web-Based Training (WBT). A self-paced stand-alone computer product used to deliver interactive subject and task knowledge.

A1.0. Career Field Education and Training Plan (CFETP)

A1.1. CFETP Purpose

- Inform and guide 3E1X1 Airmen through career progression and competency-based force development by outlining mandatory and recommended education, training and experiences.
- Provide guidance to commanders, supervisors, mentors and trainers to plan, develop, manage and conduct an efficient and effective enlisted force development program.
- Each duty location will provide unique learning opportunities to enhance Airmen career development. Annotate these learning opportunities in the individual's training record.

A1.2. Use of CFETP

- Supervisors will use this plan to ensure comprehensive and cohesive training programs are available for everyone in the 3E1X1 specialty.
- AETC training personnel will develop and revise formal resident and non-resident training based on requirements established by the users and documented in the STS. They will also work with the CFM and AFCEC Readiness and Training Division (AFCEC/COF) to develop acquisition strategies for obtaining resources needed to provide the identified training.
- Unit Training Managers (UTMs) and supervisors must ensure everyone completes the mandatory Upgrade Training (UGT) requirements specified in this plan.
- CFETP is a tool to focus the development of our Airmen Engineers to be confident and competent technicians and leaders.

A1.3. Force Development

- A joint and collaborative endeavor between the Airman, their supervisor (and/or mentor) and their commander.
- Each Airman must complete mandatory skill level upgrade and track both skill level upgrade and proficiency levels attained using their individual training records.
- Supervisors must determine appropriate levels and timing of education and training to ensure their Airmen have every opportunity to attend supplemental courses, vendor training, pursue certification, etc.
- Each Airman should develop learning plan milestones that are specific, measurable, attainable and realistic for achieving near-term (0-5 yrs), mid-term (6-10 yrs) and long-term (10+ yrs) goals and communicate these goals to their supervisors and mentors.

A1.4. Approval and Coordination

- Approval authority is the CFM under the Director of Air Force Civil Engineering at HQ USAF/A4C.
- The 3E1X1 Readiness Training Manager (RTM) will review this document annually to ensure currency and is the coordinating agency for this document.
- The 3E1X1 RTM will identify and coordinate on all AFS training requirements and updates.

A2.0. CE Occupational Badges

- Our CE badge reflects a great history and tradition, multitudes of engineers who came before us established this expectation through excellent service in both peace and war.
- When worn, Airmen will recognize you as having achieved an expected level of competence.
- Eligibility criteria for award and wear of Air Force (AF) Occupational Badges are in AFI 36-2903, *Dress and Appearance of Air Force Personnel*, Chapter 9.

A2.1. CE Badge Heraldry

- The gear wheel and compass represent the engineering profession, in both the military and in the private sector. Together, they symbolize the diverse specialties within AF Civil Engineer (CE) career fields.
- The gear represents the essence of engineering: applying scientific principles and technology to practical ends. To AF engineers, the gear symbolizes an element (representing the built environment) that meshes with others (weapon systems and trained personnel) to enable the AF to perform its mission.
- The compass is a precision tool historically used by engineers in designing and constructing facilities and equipment.



- The wings help to portray the fundamental linkage between engineering and aviation and that the built environment is the foundation supporting the AF mission and people.



2.1.2. Basic Badge Awarded upon successful completion of the apprentice course.



2.1.3. Senior Badge adds a star to the top of the badge. Wear the senior badge after award of the 7-skill level.



2.1.4. Master Badge adds a wreath around the star. Awarded to Master Sergeant or above with 5 years in the specialty from award of the 7-skill level

A3.0. Community College of the Air Force (CCAF)

- Enrollment occurs upon completion of Basic Military Training.
- Additional off-duty education is a personal choice that is encouraged for all.
- Individuals desiring to become an AETC Instructor must be actively pursuing an associate degree. A degreed faculty is necessary to maintain accreditation through the Southern Association of Colleges and Schools.
- CCAF provides the opportunity to obtain an Associate of Applied Sciences Degree. In addition to its associate degree program, CCAF offers the following:

A3.1. Air Force Credentialing Opportunities On-Line (COOL)

- CCAF manages the AF COOL Program, which provides a research tool designed to increase an enlisted Airman and Guardian's awareness of national professional credentialing and funding opportunities available for all Air Force enlisted occupational specialties. AF COOL also provides information on specific occupational specialties, civilian occupational equivalencies, specialty-related national professional credentials, credentialing agencies, and professional organizations. AF COOL includes information such as:
- Get background information about civilian credentials, including eligibility requirements and resources to prepare for an exam.
- Identify credentials relevant to an AFSC, Special Duty Identifier, and Reporting Identifier.
- Learn how to fill gaps between AF training, experience, and civilian credentialing requirements.
- Information on AF COOL funding opportunities to pay for credentialing coursework, textbooks, exams, associated fees, and recertification.
- Resources available to enlisted Airmen that can help them gain civilian job credentials.
- Credentialing requirements information on AF COOL includes information specific to enlisted AFSCs. Individuals seeking credentialing information for civilian occupations can reference the U.S. Department of Labor Website, CareerOneStop. Note: For more HVAC credentialing information, look up National Center for Construction Education & Research (NCCER) & North American Technician Excellence (NATE).

A3.2. Air University Associate to Baccalaureate Cooperative (AU-ABC)

- The AU-ABC program connects CCAF graduates with online 4-year programs. The AU-ABC program includes postsecondary institutions with institutional accreditation.
- CCAF students and graduates can search for degree programs from a list of military friendly civilian institutions via the Air Force Virtual Education Center (AFVEC). From there, students can view their education record and CCAF credits earned, apply for Military TA, and shop for baccalaureate degree programs. For more information, refer to the AU-ABC fact page.
- The program maximizes the application of military career education and training and provides a multitude of online academic and support services for enlisted members.

A3.3. CCAF Degree Completion Requirements (60 Semester Hours)

- The Mechanical & Electrical Technology Associates Degree (4VGA) applies to 3E1X1 AFSC.
- Before receiving a CCAF degree, the individual must have a 5-level and meet the following requirements:

Course	Semester Hours
Technical Education	24
Leadership, Management, and Military Studies	6
Program Elective	15
General Education Technical Education; Leadership, Management, and Military Studies; or General Education	15
Total	**60**

A3.4.1. Technical Education. (24 semester hours)

- Apply a minimum of nine (9) semester hours of CCAF institutional credit awarded from specialty-related formal training towards Technical Core subject requirements. Technical Electives are satisfied by CCAF credit and other sources in-transfer.

Technical Core Requirements	Maximum Semester Hours
AC & Refrigeration Fundamentals	6
Airfield Lighting Systems	6
Construction of Overhead	3
Electrical & Electronic Control	3
Electrical Distribution Systems	3
Electrical Fundamentals	3
Electrical Power Generation/Distribution	3
Emergency Airfield Lighting System	3
Engine System & Associated Equipment.	6
Environmental Control Systems.	9
Fuel Subsystems	3
Generator Set Operation & Aircraft Arrest Barriers	3
Heating Systems Maintenance	3
Heating Systems Operations	6
HVAC & Refrigeration Contingency.	6
HVAC/R & Civil Engineering Organization.	3
Hydrant System Maintenance	3
Launch Facility Access/Security	3
Maintenance of Aircraft Arrest System	3
Maintenance Orientation	3
Mobile Generator Set Theory & Operations	6
Power Line Equipment & Pole Climbing	3
Power Production Equipment	3
Refrigeration & AC Systems	6
Special Purpose Electrical Systems	3
Special Tools and Equipment	6
Specialized Fuel System/Tank Entry	6
Utility Fundamentals.	3
Water & Waste Distribution System	3

A3.4.2. Leadership, Management, and Military Studies (6 Semester Hours)

- CCAF accepts PME and/or civilian management courses in transfer and/or testing credit.

A3.4.3. Program Elective (15 semester hours)

- Satisfied with applicable Technical Education or General Education subjects and courses.
- Six semester hours of CCAF degree applicable technical credit otherwise not applicable to this program may be applied. See the CCAF General Catalog for details regarding the Associates of Applied Sciences degree for 3E1X1 specialty.

A3.4.4. General Education (15 Semester Hours)

General Education Subjects/Courses	Semester Hours
Communications:	
Written Communication (English Composition)	3
Oral Communication (Speech)	3
Mathematics	3
Social Science	3
Humanities	3

A3.4.6. Technical Electives

Technical Electives	Semester Hours
Air Distribution and Filtering Systems	3
Alternate Heating and Cooling	3
Blueprint Reading/Schematic Diagrams	6
Building Codes and Ordinance	3
Control Systems/Maintenance	6
CCAF Upgrade Training	15
Electricity/Electronics	9
Engine Principles	3
Environmental Awareness	3
Environmental Compliance	3
Fire-Suppression Systems	6
General Chemistry	9
General Physics	4
Hazardous Materials	6
Industrial Management	3
Industrial Safety	3
Mechanics of Soils	3
Motor, Starter, and Control Devices	6
Natural Gas Distribution	6
Quality Assurance	3
Specialty-Related Subjects In-Transfer	9
Technical Mathematics	3
Welding and Pipefitting	3

A4.0. Career Path Pyramid



A5.0. Career Path for Skill level Upgrade

ENLISTED CAREER PATH				
Education and Training Requirements	**GRADE REQUIREMENTS**			
	Rank	Average Sew-On	Earliest Sew-On	AD High Year Of Tenure (HYT) Only)
Basic Military Training School				
Apprentice Technical School (3- Skill Level) – Complete Technical School	Amn A1C	6 months 16 months		
Upgrade To Journeyman (5- Skill Level) – Complete 5-level CDC – Complete CE 5-Level WBT – Complete all 5-level core/duty related tasks – Complete WMGT 301 courses for AD and highly encouraged for the ARC – Complete AFIT WENG 170	SrA	3 years	28 months Below-the-Zone (BTZ) (22 months)	12 years
Trainer – Must be qualified and certified to perform the tasks to be trained. – Attend AF Training Course – Recommended by the supervisor				
Upgrade To Craftsman (7- Skill Level) – Minimum rank of SSgt – Complete CE 7-Level Core Concept WBT – Complete all core/duty related tasks – Complete AFIT WENG 200 and WENG 270 courses for AD and highly encouraged for the ARC – Complete AFIT WMGT 322 and WENG 370	SSgt TSgt	5 years 9 years	3 years 5 years	22 years 24 years
Certifier – SSgt with 5-skill level or civilian equivalent – Attend AF Training Course – Appointed by the commander – Be someone other than the trainer (for core and critical tasks only)				
Upgrade To Superintendent (9- Skill Level) – Minimum rank of SMSgt – In-residence USAF SNCOA, or sister service equivalent, graduation is required for SMSgt sew-on (AD only) – Completion of Troop Construction Project Management Course (AFIT WMGT 437) is mandatory for Active Duty. This course is highly recommended for AD & ARC MSgts	SMSgt	20 years	11 years	28 years
SEL – Chief Orientation Course (AFR Only) – Completing the Civil Engineer Superintendent Course (AFIT WMGT 570) is mandatory for Active Duty, ANG, and AFRC. Note: This is not a skill-level-awarding course. – In-residence Chief Leadership Course is required for all CMSgt selects and CMSgts prior to 1-year time in grade	CMSgt	22 years	14 years	30 years

A6.0. Enlisted Professional Military Education (EPME)

A6.1. Basic EPME (Distance Learning)

- Basic EPME requirements are completed via distance learning (DL) courses to establish a foundation for continued development and include the Noncommissioned Officer (NCO) DL and Senior Noncommissioned Officer (SNCO) DL courses.

A6.2. Resident EPME (In-residence)

- Resident EPME requirements include Airman Leadership School (ALS), NCOA, SNCOA and the Chief Leadership Course (CLC).
- Resident EPME completion is required for Active Duty AF promotion to the grades of SSgt, MSgt and SMSgt. See AFI 36-2502, *Enlisted Airman Promotion and Demotion Programs*.

A6.3. Resident EPME Eligibility Chart

EPME Course	Selection Priority
ALS	1. E-5 with approved waivers and E-5-selects 2. E-4 with the most TIG and a pending assignment or deployment ahead of other service members 3. Eligible DAF and DoD and other civilians, sister service E-4, approved international students, and local civilians
NCOA	1. E-7 with approved waivers 2. E-7 selects 3. E-6 (non-selects) by seniority (based on date of rank and total active federal military service date) 4. Eligible DAF and DoD civilians, sister service and international students per the Program Requirement Document
SNCOA	1. E-8 with approved waivers 2. E-8 selects 3. E-7 (non-selects) by seniority (based on date of rank and total active federal military service date) 4. Eligible DAF and DoD civilians, sister service and international students per the Program Requirement Document
CLC	1. E-9 by seniority (based on date of rank and total active federal military service date) 2. E9 selects 3. Eligible DAF and DoD civilians, sister service and international students

B1.0. Competency Based Learning (CBL)

B1.1. Origin

- AETC rebuilt our entire training platform by using competencies to identify the behaviors needed to be successful on the job. AETC accomplished this by working with 3E1X1 SMEs to identify Occupational Competencies and link them to the Airman's Foundational Competencies in a manner that provides all 3E1X1 Airmen with an unbiased learning path to become a confident and competent Airman Engineer.

B1.2. Purpose

- A competency-based learning system provides the 3E1X1 community with a unifying strategy towards force development. Each HVAC Airmen will have a clear objective and individualized training plan to sharpen their tactical expertise, see STS in Part II.
- A competency-based learning system will allow each Airmen to move beyond the minimum 3-, 5-, 7- & 9- skill level training requirements.
- A competency-based learning system allows the 3E1X1 community to transition our education and training plan to a student-centered approach so that each Airman can manage their own professional growth by identifying their own strengths and weaknesses.

B1.3. Use

- The foundations of a successful education and training plan are it must be clear, concise and specific. Our Occupational Competencies (Figure 1) are a set of competencies that our CE leaders value and are the core of this education and training plan.



Figure 1: 3E1X1, HVAC/R, Competency Wheel

B2.0. Airman's Foundational Competencies

- The Airman's Foundational Competencies (Figure 2) are those competencies that our AF leaders value and are universally applicable to all Airmen.
- These competencies are at the core of Airmen development and enable each Airman with tools, pathways and capabilities to improve their performance on the job and in any situation.
- The four Airman's Foundational Competency domains are Developing Self, Developing Others, Developing Organization, and Developing Ideas.



Figure 2: Airman's Foundational Competencies

B2.1. Airman's Foundational Competency Self-Assessment

- Each Airman can go to MyVECTOR to complete a self-assessment, which will have them evaluate themselves on the 23 Airman's Foundational Competencies. Additionally, 360° surveys are available where subordinates, peers and leaders can also provide feedback.
- These self-assessment tools will provide each Airman with immediate feedback on personal strengths and areas for improvement.
- Each Airman will receive a personal improvement plan with targeted resources (videos, reading content, developmental opportunities) for their continued self-development.

B3.0. Occupational Competency Model

- A career field's competencies can be viewed in a competency model, which is an organized collection of competencies pertinent to the career field. The 3E1X1 Occupational Competency Model (Figure 3) provides a framework to develop/build an efficient and effective training program.
- This model provides each Airman with an overview of all competencies and sub-competencies required in the 3E1X1 career field.
- This model allows each Airman to see how their OJT, AETC courses, education, other training and experiences align with the overall career field's strategic objective.

Note: The descriptions used are 3E1X1 career field's translation of these sub-competencies.

Competency	Sub-Competency	Description
Management	Programs	A set of measures or activities that follow a set of temporary or continuously changing guidelines.
	Supervision	Coordinates and manages tasks and leads individuals and/or sections' collective efforts to accomplish organizational goals.
	Work center Management	Manages work center task requirements to include labor, logistics, and process management actions required to complete objectives of leadership and organization.
	Training	Actions to develop and increase Knowledge, Skills, Abilities, and Other Characteristics (KSAOs) enabling successful execution of tasks to support mission requirements.
Install, Maintenance, and Operations	Electrical	Flow of electrons between positive and negative points with conductors used to power HVAC/R and control systems for a customer or process.
	Hydronics	The use of water as a medium through piping and terminal units to transfer heat in cooling or heating systems.
	Cooling	The process of removing heat from the conditioned space through the utilization of various types of environmental control units.
	Heating	The process of adding heat to a conditioned space through the utilization of radiation, convection and conduction.
	Control Systems	Equipment is used to maintain, monitor and regulate energy consumption of a building by controlling the operation of energy consuming HVAC/R systems.
	Air Distribution/Ventilation	Distribution and ventilation of air to and from a space using fans, filters, ductwork, dampers, etc.
Safety	Safety	To remain vigilant to one's safety, safety of others, and equipment. To recognize hazards associated with a task or environment. To be cognizant of the inherent dangers of bodily harm, injury, death, or damage to equipment.

Figure 3: 3E1X1, HVAC/R, Occupational Competency Model

B4.0. Occupational Competency Rubric

- Competency rubrics will guide each Airman through training and development opportunities by synchronizing formal and informal training efforts with our career field's strategic objectives.
- Competency rubrics consist of supporting competencies, proficiency levels and observable behaviors. Provided in Figure 4 is our electrical competency rubric, and it is used here as an example. All 3E1X1 competency rubrics are embedded in the STS in Part II.
- Observable behaviors are professional development expectations that provide objective evidence Airmen have mastered the necessary proficiency levels needed for successful job performance.
- Our top 10 supporting competencies listed across all 3E1X1 competencies are troubleshooting, self-awareness, verbal communication, problem-solving, design sense, mentoring, accountability, technology savvy and time management.

  Note: There is no time requirements associated with obtaining each proficiency level. Airmen Engineers must not progress until they have demonstrated mastery of each proficiency level, starting with Basic.

<table>
<tr><th>Competency</th><th>Proficiency Levels</th><th>Observable Behaviors</th></tr>
<tr><td>Install, Maintenance and Operation

Sub-Competency</td><td>Expert

Consistency of Application: Able to innovate and formulate strategies; able to model/guide/teach others the competency of how to apply the competency</td><td>- Designs HVAC/R electrical systems in coordination with project manager and engineer
- Provides mechanical and electrical expertise to project managers/engineers for modification and development of HVAC/R electrical systems</td></tr>
<tr><td rowspan="2">Electrical

Description

Flow of electrons between positive & negative points using conductors used to power HVAC/R & control systems for customer or process</td><td>Advanced

Consistency of Application: Sustained application of competency over time in complex situations</td><td>- Modifies HVAC/R electrical systems IAW regulations (e.g. NEC, NFPA, etc.)
- Integrates electrical system with control system (CS) and verify system operation with CS technicians
- Performs advanced electrical troubleshooting (solid-state components, PLC, DC voltage, frequency drive, etc.)</td></tr>
<tr><td>Intermediate

Consistency of Application: Sustained application of competency over time in a variety of situations</td><td>- Identifies and troubleshoot electrical system malfunctions
- Replaces electrical system components and verify system operations
- Installs electrical distribution and components based on interpretation of schematic & sequence of operation</td></tr>
<tr><td>Supporting Competencies

Troubleshooting, Cautious, Critical Thinking, and Problem Solving</td><td>Basic

Consistency of Application: Sustained application of competency over time</td><td>- Assists technician with electrical system tasks such as installing, repairing and replacement
- Performs minor preventive maintenance action (proper electrical connections, verified applied system voltage, amperage draws, capacitor reading, etc.)</td></tr>
</table>

Figure 4: 3E1X1, HVAC/R, Electrical, Competency Rubric

B5.0. Occupational Competency Map

- The 3E1X1 Competency map allows each Airman and their supervisors to gain a clear and objective view of the expected proficiency levels for a specific duty position.
- This competency map is a measuring tool to determine if an Airman has achieved mastery in a specific or similar duty position. In addition, this tool will allow Airman to prepare for a future duty position.
- The 3E1X1 competency map provided in figure 5 is comprised of our Occupational Competencies and common duty positions found in the 3E1X1 career field.
- The numbers used on the competency map are associated with the four proficiency levels found on a competency rubric **(1=Basic, 2=Intermediate, 3=Advanced and 4=Expert).**

Note: Each duty location provides unique learning opportunities and Airmen may not be able to achieve certain proficiency levels at their current duty location.

HVAC/R	Helper	Technician	Controls Technician	Craftsman	Shop PM Program Manager	Training instructor / Cadre	R&O SME	Shop NCOIC	LE / Crse Developer	Superintendent	RTM / Staff
Management											
Programs	N/A	1	2	2	3	2	2	3	2	3	4
Supervision	N/A	1	1	1	2	1	2	4	3	4	4
Work Center Management	N/A	1	1	1	2	1	2	3	1	4	4
Training	N/A	1	1	2	3	3	3	4	3	4	4
Install, Maintain & Operate											
Electrical	1	2	2	3	2	3	4	3	3	3	3
Hydronics	1	2	2	3	3	3	4	3	3	3	3
Cooling	1	2	2	3	2	3	4	3	3	3	3
Heating	1	2	2	3	3	3	4	3	3	3	3
Control Systems	1	2	3	3	2	3	4	3	3	3	3
Air Distribution/Ventilation	1	2	2	3	2	3	4	3	3	3	3
Safety											
Safety	1	2	2	2	3	4	2	4	2	4	2

Figure 5: 3E1X1, HVAC/R, Competency Map

Section C – Mandatory Skill Level Training Requirements

C1.0. Skills and Career Progression. Adequate training and timely progression from the apprentice to the superintendent level plays an important role in the Air Force's ability to accomplish its mission. It is essential that everyone involved in training must do their part to plan, manage and conduct an effective training program. The guidance provided in this part of the CFETP ensures individuals receive viable training at the appropriate point in their career. Listed below are all the mandatory and recommended training requirements:

C1.1. Initials Skills Training. Mandatory training requirements. The STRT members reviewed and updated the initial skills course content; see the STS in Attachment 2, for the additions, deletions and modifications made to this course.

C1.2. 5-Level Upgrade Training Requirements. Mandatory training requirements. The STRT members reviewed, validated, and updated existing and future CDCs requirements to ensure only current material remained and added new technology information.

C1.3. 7-Level Upgrade Training Requirements. Mandatory training requirements. The STRT members reviewed, validated, and updated STS line items. Completion of the 10-day in-residence Craftsman Course at Sheppard AFB, TX is required for upgrade.

C1.4. Continuation training. If needed, supervisors must develop a continuation-training program that ensures personnel in the 3E1X1 career field receive the minimum necessary training at the appropriate points in their careers. Any additional shop/unit specific knowledge and skill requirements not taught through initial skills or upgrade training is continuation training. The purpose of continuation training is to provide training that exceeds minimum mandatory training requirements with emphasis on present and future duty positions at a specific duty location.

C1.5. Supplemental Courses. The purpose of these courses is to enhance our upgrade training. Additionally, if an Airman is transitioning from one extreme climate to another, they should attend these supplemental course(s) before PCSing or deploying. Work with your supervisor and UTM to schedule these courses.

C1.6. Competency Based Learning System. Competency development is self-paced learning that allows Airmen to move beyond the minimum mandatory skill level training requirements. For example, 3E1X1 Airmen can obtain their 7-skill level and still be at the basic proficiency level at some duty locations. This allows our Airmen to address developmental gaps and strengthen their capabilities after completing their minimum skill level requirements.

C2. Mandatory Training Sequence of Activities

C2.1. Purpose. This section outlines the 3E1X1 specialty training activities and sequence for award to 3-, 5-, 7- and 9- skill levels.

C2.2. Specialty Qualification. Airmen must complete all skill level requirements before they can progress to the next skill level/training requirement.

C2.2.1. Apprentice (AFSC 3E131 – AB, AMN, A1C).

C.2.2.1.1 Complete initial skills training at Sheppard AFB.

C2.2.2. Journeyman (AFSC 3E151 – SrA, SSgt).

C2.2.2.1. Complete Civil Engineer (CE) 5-Level Core Concepts Course Distance Learning (DL) product located on the **myLearning** prior to beginning CDCs and DL courses.

C2.2.2.2. Complete Air Force Qualification Training Packages (AFQTPs) with a trainer and complete web-based courses to enhance knowledge in this career field.

C2.2.2.3. Complete 5- level CDCs, Core and Diamond task requirements.

C2.2.2.4.Completing WMGT 301 Intro to Asset Management course is mandatory for Active Duty and highly encouraged for the Air Reserve Component. Completing WENG 170 Cybersecurity for Control Systems is mandatory for Total Force.

C2.2.2.5. If required by unit or workcenter, complete continuation training to broaden technical knowledge, skill, and experience.

C2.2.2.6. After all, 5- skill level requirements are completed, supervisors and UTMs must coordinate upgrade procedures.

C2.2.2.7. Use STS in Attachment 2 to create learning goals to enhance competence.

C2.2.2.8. Pursue a Community College of the Air Force (CCAF) degree.

C2.2.2.9. A 5- skill level (journeyman) may be assigned job positions such as team leader, shift supervisor and task trainer.

C2.2.2.10. Use CDCs and other reference material to prepare for Weighted Airman Promotion System (WAPS) testing.

C2.2.3. Craftsman (AFSC 3E171 – SSgt, TSgt, MSgt).

C2.2.3.1. Complete CE 7-Level Core Concepts Course located on **myLearning**

C2.2.3.2. Complete all Core and Diamond tasks.

C2.2.3.3. Completing AFIT WENG 200 Scoping and Estimating is mandatory for Active Duty and highly encouraged for the Air Reserve Component. Completing WMGT 322 Intro to Project Management, and WENG 370 Control Systems Cybersecurity for CE Leaders courses is mandatory for the Total Force.

C2.2.3.4. Complete the HVAC/R Craftsman Course. **Note:** Members attending this course must have completed all OJT requirements for 7-level upgrade prior to attending this course.

C2.2.3.5. If required by unit or workcenter, complete continuation training to broaden technical knowledge, management skills and experience.

C2.2.3.6. After all, 7- skill level requirements are completed, supervisors and UTMs must coordinate upgrade procedures.

C2.2.3.7. Complete Resident EPME as outlined in *DAFI 36-2685, Total Enlisted Developmental Education*.

C2.2.3.8. Continue academic education through CCAF and higher degree programs.

C2.2.3.9. Attend AETC supplemental courses and vendor training to enhance competence.

C2.2.2.10. Completing of the AFIT WENG 270 Advanced Control Systems Cybersecurity Course is mandatory for Active Duty and required for promotion to TSgt. This course is highly encouraged for Air Reserve Component TSgts.

C2.2.3.11. Use CDCs and other reference material to prepare for WAPS testing. **Note:** Airmen eligible for promotion to the rank of E-7 through E-9 are not required to WAPS test.

C2.2.4. Superintendent (AFSC 3E490 - SMSgt)

C2.2.4.1. Must be a SMSgt for award to the 9- skill level.

C2.2.4.2. Completing the Troop Construction Project Management Course (AFIT WMGT 437) is mandatory for Active-Duty personnel. This course is highly recommended for the Active Duty and Air Reserve Component MSgts.

C2.2.4.3. Should take continuation training course to increase knowledge of budget, manpower, resources, and personnel management.

C2.2.4.4. Must complete the appropriate Basic EPME and Resident EPME as outlined in *DAFI 36-2685, Total Enlisted Developmental Education.*

C2.2.4.5. Continue academic development through higher education is recommended.

C2.2.5. Senior Enlisted Leader (SEL) (3E000 - CMSgt).

C2.2.5.1. Must be selected for CMSgt and possess qualifications in a feeder specialty (3E090, 3E290, 3E490, 3E591 or 3E691).

C2.2.5.2. Completing the Civil Engineer Superintendent Course (AFIT WMGT 570) is mandatory for Active Duty, ANG, and AFRC. **Note:** This is not a skill-level-awarding course.

C2.2.5.3. SELs work in multiple leadership positions and functional areas that challenge them and effectively use their general managerial and supervisory abilities.

C2.2.5.4. Must complete the appropriate Basic EPME and Resident EPME as outlined in *DAFI 36-2685, Total Enlisted Developmental Education*.

SECTION C3 - SKILL LEVEL TRAINING REQUIREMENTS

C3.1. Purpose. This section outlines the 3E1X1 specialty qualification requirements for each skill level and establishes the mandatory requirements for each skill level's entry, award, and retention. The STS in Attachment 2 identifies the core tasks, diamond tasks, and skill level requirements.

C3.2. Specialty Qualification. Refer to **myFSS** for the most current minimum requirements for entry into the 3E1X1 career field. See DAFECD Attachment 4 for additional entry physical requirements.

C3.2.1. Apprentice (3-Level) Training Requirements. (3E131)

KNOWLEDGE	Completion of the HVAC/R Apprentice course pipeline, J8AQR3E131 00AB ITRO and J3ABR3E131 00AD AF Unique, at Sheppard AFB, TX.
EDUCATION	For entry into this specialty, completion of high school with courses in mathematics, general science, physics, shop mechanics, electricity, and computer fundamentals are desirable.
TRAINING	Completion of the HVAC/R Apprentice course is mandatory for award of this skill level.
EXPERIENCE	None required.
OTHER	Maintain local network access IAW AFI 17-130, Cybersecurity Program Management and AFMAN 17-1301, Computer Security (COMPUSEC). For award and retention of AFSC 3E131, Airmen must possess a valid state driver's license to operate GMV IAW AFI 24-301, Vehicle Operations. Normal color vision as defined in DAFMAN 48-123, Medical Examinations and Standards.
IMPLEMENTATION	The Unit Training Manager at member's gaining unit of assignment submits for award of the 3-skill level upon graduation of the apprentice course.
TRAINING SOURCES & RESOURCES	Reference PART II, SECTIONS: B - COURSE OBJECTIVE LIST (COL); C - SUPPORT MATERIAL; D – EDUCATION AND TRAINING COURSE INDEX; E – MAJCOM UNIQUE REQUIREMENTS; F - HOME STATION TRAINING.

C3.2.2. Journeyman (5-Level) Training Qualifications. (3E151)

KNOWLEDGE	Must have knowledge on: Principles of HVAC/R systems, controls, and components; combustion systems; air and water balancing; industrial air compressors; interpreting drawings and schematics, fundamentals of HVAC/R equipment and troubleshooting techniques; water treatment; environmental regulations for fuels, refrigerants, and hazardous materials; military and commercial publications; and environmental regulations.
EDUCATION	**The following education is desirable and strongly encouraged:** CCAF Mechanical & Electrical Technology Associates Degree (4VGA).
TRAINING	Completion of 5-level CDCs. Completion of all the paper-based AFQTPs and their associated web-based courses on **myLearning** for all core (5^) and diamond (♦) tasks with a minimum passing score of 80%. Completion of the CE 5-Level Common Core Concept web-based course located on **myLearning** is mandatory for award of this skill level. Completing WMGT 301 Intro to Asset Management course is mandatory for Active Duty and highly encouraged for the Air Reserve Component. Completing WENG 170 Cybersecurity for Control Systems is mandatory for Total Force. Certification of all 5-skill level core tasks identified with a (5^) in the task column of the STS. Certification of 5-skill level diamond tasks identified with a diamond (♦) in the task column, if the equipment is available. Minimum requirement is completing the AFQTP mandated web-based course. Certification of duty position requirements as identified by the supervisor. **The following training is desirable and strongly encouraged:** AFIMSC Centralized Vendor Training Opportunities.
EXPERIENCE	Qualification in and possession of AFSC 3E131. Technical proficiency in operation and maintenance of HVAC/R systems and process; electrical troubleshooting and assigned contingency equipment.
OTHER	Maintain local network access IAW AFI 17-130, Cybersecurity Program Management and AFMAN 17-1301, Computer Security (COMPUSEC). For award and retention of AFSC 3E151, Airmen must possess a valid state driver's license to operate GMV IAW AFI 24-301, Vehicle Operations. Normal color vision as defined in DAFMAN 48-123, Medical Examinations and Standards.
TRAINING SOURCES & RESOURCES	Reference PART II, SECTIONS: B - COURSE OBJECTIVE LIST (COL); C - SUPPORT MATERIAL; D – EDUCATION AND TRAINING COURSE INDEX; E – MAJCOM UNIQUE REQUIREMENTS; F - HOME STATION TRAINING.

IMPLEMENTATION	Initiate entry into 5-level upgrade training after the individual has completed all 3-level requirements. Initiate Qualification Training any time individuals cannot perform their new duties. Use OJT, CDCs, AFJQSs, and AFQTPs concurrently to obtain the necessary qualifications.

C3.2.3. Craftsman (7-Level) Training Requirements. (3E171)

KNOWLEDGE	All 3- and 5-level knowledge requirements apply to 7-level.
EDUCATION	**The following training is desirable and strongly encouraged:** CCAF Mechanical & Electrical Technology Associates Degree (4VGA). Higher education through a civilian institution.
TRAINING	Completion of all the paper-based AFQTPs and their associated web-based courses on **myLearning** for all 7-skill level core (7^) and diamond (♦) tasks with a minimum passing score of 80%. Completion of the CE 7-Level Common Core Concept web-based course located on **myLearning** is mandatory for award of this skill level. Certification of duty position requirements as identified by the supervisor. Completing AFIT WENG 200 Scoping and Estimating is mandatory for Active Duty and highly encouraged for the ARC. Completing WMGT 322, Intro to Project Management, and the WENG 370 Control Systems Cybersecurity for CE Leaders courses is mandatory for the Total Force. Completion of AFIT WENG 270 Advanced Control Systems Cybersecurity Course is mandatory for Active Duty and required for promotion to TSgt. This course is highly encouraged for ARC TSgts. Certification of all 7-skill level diamond tasks identified with a diamond (♦) in the task column of the STS if the equipment is available. Minimum requirement is completing the AFQTP mandated web-based course. Completion of the 10-day Craftsman Course (J3ACR3E171 00AC) for award of the 7-skill level is required. Members attending this course must have completed all 7-level upgrade requirements prior to attending this course. **The following training when available is strongly encouraged:** Completion of the J3AZR3E151 00RA (5 days), in residence, Advanced HVAC/R Cooling and Refrigeration Systems Course at Sheppard AFB TX. Completion of the J3AZR3E151 00HA (5 days), in residence, Advanced HVAC/R Heating Systems Course at Sheppard AFB, TX. Completion of the J3AZR3E151 00CA (5 days), in residence, Advanced HVAC/R Building Automation Systems Course at Sheppard AFB, TX. Completion of the J3AZR3E151 00BA (5 days), in residence, Advanced HVAC/R Air and Hydronic Balancing Course at Sheppard AFB, TX.

EXPERIENCE	Qualification in and possession of AFSC 3E151. Performing or supervising functions of HVAC/R systems, new facility design review and approval, psychrometric procedures, calculating heat lead load requirements, measuring subcooling and superheat, and electrical troubleshooting.
OTHER	HVAC/R craftsmen should pursue additional knowledge and skills not taught through initial skills or upgrade training. The purpose of ongoing training is to exceed minimum upgrade requirements with emphasis on HVAC/R Airmen achieving the necessary training and experience at the appropriate point in their career to be more effective in present and future duty positions. Maintain local network access IAW AFI 17-130, Cybersecurity Program Management and AFMAN 17-1301, Computer Security (COMPUSEC). For award and retention of AFSC 3E171, Airmen must possess a valid state driver's license to operate GMV IAW AFI 24-301, Vehicle Operations. Normal color vision as defined in DAFMAN 48-123, Medical Examinations and Standards.
TRAINING SOURCES & RESOURCES	Reference PART II, SECTIONS: B - COURSE OBJECTIVE LIST (COL); C - SUPPORT MATERIAL; D – EDUCATION AND TRAINING COURSE INDEX; E – MAJCOM UNIQUE REQUIREMENTS; F - HOME STATION TRAINING
IMPLEMENTATION	Initiate entry into 7-level upgrade training after the individual has completed all 5-level requirements and selected for SSgt. Initiate Qualification Training any time individuals cannot perform their new duties. Use OJT, CDCs, AFJQSs, and AFQTPs concurrently to obtain the necessary qualifications.

C3.2.4. Superintendent (9-Level) Training Requirements. (3E490)

KNOWLEDGE	Principles of water distribution and wastewater collection systems, water and wastewater treatment systems, plumbing systems, natural gas distribution systems, liquid fuels distribution systems, pest management activities, controls, and components, combustion systems, air and water balancing, non-electric kitchen equipment, industrial air compressors, and backflow prevention devices. Interprets drawings and schematics, industrial wastewater treatment, and usage of maintenance materials such as pipes, valves, fittings, packing and gaskets, electrical and electronic controls that apply to these systems, wiring schematics and diagrams, military and commercial publications, environmental regulations for fuels, pesticides, refrigerants, and hazardous materials.

EDUCATION	Completion of the grade appropriate EPME is mandatory. **The following training is desirable and strongly encouraged:** CCAF Mechanical & Electrical Technology Associates Degree (4VGA). Higher education through a civilian institution.
TRAINING	Completing the Troop Construction Project Management Course (AFIT WMGT 437) is mandatory for Active-Duty personnel. This course is highly recommended for the Air Reserve Component.
EXPERIENCE	For award of the AFSC 3E490 qualification in and possession of AFSC 3E171, 3E471 or 3E473 is mandatory. Must be a SMSgt. Experience in Civil Engineer management functions such as operations, facilities system maintenance, repair of distribution and collection systems, natural gas distribution systems, heating, ventilation, cooling and refrigeration systems, liquid fuels distribution systems, pest management, and industrial control systems.
TRAINING SOURCES & RESOURCES	Reference PART II, SECTIONS: B - COURSE OBJECTIVE LIST (COL); C - SUPPORT MATERIAL; D – EDUCATION AND TRAINING COURSE INDEX; E – MAJCOM UNIQUE REQUIREMENTS; F - HOME STATION TRAINING.
OTHER	Infrastructure Systems Superintendents should pursue any additional knowledge and skill requirements not taught through initial skills or upgrade training. The purpose of ongoing training is to exceed minimum upgrade requirements with emphasis on personnel achieving the necessary training and experience at the appropriate point in their career to be more effective in present and future duty positions. Recommended areas of study include but are not limited to AFIT Training courses, such as Project and Contract Management courses. Maintain local network access IAW AFI 17-130, Cybersecurity Program Management and AFMAN 17-1301, Computer Security (COMPUSEC). For award and retention of AFSC 3E490, Airmen must possess a valid state driver's license to operate GMV IAW AFI 24-301, Vehicle Operations. Normal color vision as defined in DAFMAN 48-123, Medical Examinations and Standards.
IMPLEMENTATION	Entry into 9-level training is initiated when an individual is selected for SMSgt and is a fully qualified 7-Level. Qualification training is initiated any time an individual is assigned duties they are not certified to perform.

C3.2.5.SEL. (3E000)

KNOWLEDGE	Knowledge is mandatory of managing and directing personnel resource activities, interpreting, and enforcing policy and applicable directives, establishing control procedures to meet work goals and standards, recommending, or initiating actions to improve operational efficiency, planning and programming work commitments and schedules, developing plans regarding facilities, supplies, and equipment procurement and maintenance.
TRAINING	Reserve Component Chief Orientation Course (AFRC only). Completing the Civil Engineer Superintendent Course (AFIT WMGT 570) is mandatory for Active Duty, ANG, and AFRC. **Note**: This is not a skill-level-awarding course.
EXPERIENCE	Possess qualifications in feeder specialty prior to award of Civil Engineer Manger code 3E000. Managerial ability to plan, direct, coordinate, implement, and control a wide range of work activity.
EDUCATION	Completion of the grade appropriate EPME is mandatory.
OTHER	N/A.
IMPLEMENTATION	Selection to CMSgt enters an individual into the Civil Engineer Manager Code 3E000. They must possess the qualifications from a feeder specialty (3E090, 3E290, 3E490, 3E591, and 3E691).

Section D – Resource Constraints

D1. Purpose This section identifies known resource constraints that preclude optimal training at the schoolhouse, including information such as cost and manpower. Explanations of each resource constraint and what effect each constraint has on training are included. Also included in this section are actions required, Office of Primary Responsibility (OPR) and target completion dates. If applicable, resource constraints will be, at a minimum, reviewed and updated annually.

D2. **Apprentice (3-Level) Training:**

D2.1. Constraints. None.

D3. **Journeyman (5-Level) Training:** Minimal.

D3.1. Constraints. New web-based CDC is in development.

D3.1.1. Impact. Graduating Airmen will utilize current set until new web-based CDC is available.

D3.1.2. Resources Required. None.

D3.1.3. Action Required. Complete development and post on myLearning.

D3.2. OPR/Target Completion Date. Learning Engineer/To be determined (TBD).

D4. Craftsman (7-Level) Training: Minimal.

D4.1. Constraints. New Craftsman Course is under development.

D4.1.1. Impact. Airman in upgrade to attain their 7-level will take current craftsman course.

D4.1.2. Equipment Constraints. None.

D4.2. OPR/Target Completion Date. 366 TRS Training Manager/June 2026.

D5. Superintendent (9-Level) Training: Not Applicable.

Section E – TRANSITIONAL TRAINING GUIDE

E1. Purpose. This transition guide will provide the direction and guidance to prepare our Airmen for the 21st century. To be successful tomorrow, plan today to assure our strength and direction.

E1.1. Supervisors must be qualified on the training requirement before they can sign off a trainee's records. Additionally, they must complete the Air Force Training Course before signing off trainees on any training requirement.

E1.2. For core & diamond tasks use the AFQTPs on **myLearning** or **CE DASH** under the documents tab in the AFSC AFQTP folder.

E1.3. For transitional training guide questions, please contact the AFCEC/COF Training Support Section at DSN 523-6879 or comm. 850-283-6879 or email **afcec.ce.training@us.af.mil**.

Airmen assigned to the Requirements and Optimization (R&O) section must complete all training and comply with all tasks defined in the R&O Master Task List located on **myLearning**.

PART II

SECTION A – SPECIALITY TRAINING STANDARD

A1. Implementation. Use this STS for technical training provided by AETC for 3-level classes beginning in June 2025 and for the 7- level Craftsman course beginning January 2026.

A2. Purpose. As prescribed in DAFMAN 36-2689, *Training Program* and in collaboration with the Air Force CFM, it is mandatory for all civil engineers, regardless of duty assignment, to use the paper-based STS or an automated training record, when available.

A2.1. **Column 1 (*Tasks, Knowledge, and Technical References*).** Lists the most common tasks, knowledge, and supporting technical references (TRs) necessary for Airmen to perform duties in the 3-, 5-, and 7-skill level.

A2.1.1. **Task Qualification Tasks (TQT).** In accordance with DAFI 10-2503, *Chemical, Biological, Radiological, Nuclear (CBRN) Defense Program*. TQT requirements identified by (TQT) after the line item of the STS are mandatory wartime skills that Airmen will perform while wearing Individual Protective Equipment.

A2.2. **Column 2 (*Core Tasks*).** Column 2 identifies core tasks (specialty-wide training requirements) by a number (5^ or 7^) in the skill level column. **As a minimum, trainees must complete all core and critical tasks for skill-level upgrades.**

A2.2.1. **Wartime Tasks.** In response to a wartime posture, all tasks will be taught in the 3-level course at an accelerated rate in a streamlined environment. A "Wartime Training Posture" will be 6-days a week/10-hour day.

A2.2.2. **Diamond Tasks.** Tasks (column 1) identified by a diamond (♦) after the task item are considered contingency/war tasks and are critical to the career field. Equipment shortfalls at most locations have created problems with the actual hands-on certification of these tasks. When required equipment is unavailable for instruction, completion of the corresponding task AFQTP is required for upgrade/qualification training.

A2.3. **Column 3 (*Certification for OJT*).** Used to record completion of tasks and knowledge training requirements. Use paper or automated training management applications to document technician qualifications. **Task certification of core and critical tasks** requires a training completion date and the initials of the trainee, trainer, and certifier. All non-core tasks require the training completion date and initials of the trainee and trainer only.

A2.4. **Column 4 (*Behavioral Codes Used to Indicate Training/Information Provided*).** Indicates formal training and correspondence course requirements. It shows the proficiency to be demonstrated on the job by the graduate because of training on the task, knowledge, and career knowledge provided by formal courses and correspondence courses. See CADRE/AFSC/CDC listing maintained by the UTM for current CDC listings.

A2.5. **Qualitative Requirements.** This CFETP contains a Competency Based Learning Code Key used to indicate the level of training and knowledge provided by resident training, and career development courses.

A2.6. **Job Qualification Standard (JQS).** The STS becomes an AFJQS for OJT in an automated training application and is used according to DAFMAN 36-2689, *Training Program*. Supervisors and trainers use AFQTPs to ensure Air Force-wide standardized procedures for training core tasks. When used as a JQS, the following requirements apply:

A2.6.1. **Documentation.** Document and certify completion of training:

A2.6.1.1. **Duty position.** Duty position requirements will be developed and identified by the work center supervisor, maintained in the shop, or loaded into the automated training management application when available. Completing core, critical, and diamond tasks is mandatory for all duty positions. Work center supervisors assign Airmen to the correct duty position.

A2.6.1.2. **AFQTP Training and Documentation.** AFQTPs or AFQTP assessments for all core (5^ or 7^) tasks and their completion are mandatory to fulfill task knowledge requirements for upgrade/qualification training. Each AFQTP provides the step–by–step procedures for the trainee, trainer, and certifier in completing each core or diamond task and instructions on how to document the training in the individual training record.

A2.6.1.2.1. **Training.** Required documentation of the start and completion of the AFQTP and the training record for all core tasks. Diamond tasks require the completion of the web-based course (with the review and post-test located in the program) or completing the AFQTP assessment on **myLearning** to determine if the trainee has attained the required knowledge level. Once the trainee has completed the web-based course or AFQTP assessment, Airmen provide the course completion certificate to the trainer/supervisor for annotation in the training record.

A2.6.1.2.2. **Hands-On Training.** *DO NOT sign off the tasks in the JQS until the trainee has completed hands-on/certification training.* For diamond tasks, if the equipment is not available at home station, completing the AFQTP or AFQTP assessment is the ONLY requirement for upgrade. Sign off the trainee's paper JQS or automated training record when the trainee receives training on the equipment at home station or at a TDY location.

A2.6.2. **Transcribing from previous versions to new CFETP.** Transcribing documentation to a new CFETP is an administrative function, not a re-evaluation of training. Therefore, supervisor and trainer are considered synonymous with documentation. Transcribe within 120 calendar days (240 calendar days for ARC) of the CFETP revision date or from the date revision is posted to the automated training records system.

A2.6.2.1. **Previous training certification not listed.** Place an entry into the trainee DAF Form 623A; the transcriber and trainee must acknowledge entry.

A2.6.2.2. **Transcribing external training certification.** If a trainee attended a formal training course and received appropriate accreditation, place an entry into the trainee's DAF Form 623A.

A2.6.3. **Documenting Career Knowledge.** When a CDC course is not available, the supervisor identifies STS training references that the trainee requires for career knowledge IAW DAFMAN 36-2689, *Training Program* and ensures, as a minimum, that trainees cover all mandatory items specified in AFMAN 36-2100, *Military Utilization and Classification*. For two-time CDC course exam failures, the unit commander will take appropriate action IAW DAFMAN 36-2689, *Training Program*. Document career knowledge to submit a CDC course waiver.

A2.6.4. **Decertification.** When a supervisor determines an Airman is unqualified on a task previously certified for their duty position, the supervisor erases the previous certification or deletes the certification when using an automated system. Enter appropriate remarks pertaining to the reason for decertification on the DAF Form 623A or the automated version. Once an Airman has shown proficiency, the trainer and certifier can sign off the task(s).

A2.6.5. **Recertification.** Supervisors use the DAF Form 1098 to document tasks requiring recurring training or evaluation. Air Force directives may identify tasks in the CFETP requiring special certification or evaluation. This form may be printed and filed in the DAF Form 623. This form will remain in the training record for tasks requiring certification until superseded or no longer required.

A2.6.6. **Training Standard.** Train and certify Airman to the "go" level. Go means the individual can perform the task without assistance and meets the local requirements for accuracy, timeliness, and correct use of procedures. Use available AFQTPs to train tasks.

A2.7. **Specialty Training Standard.** The STS is a guide for developing promotion tests used in the Weighted Airman Promotion System (WAPS). Senior NCOs with extensive practical experience in their career fields develop Specialty Knowledge Tests (SKTs) at the AETC Airman Advancement Division. Subject matter experts authenticate WAPS material and reference AF Specialty-specific occupational analysis data to develop SKTs. They develop questions based on the study references listed in the Enlisted Promotions References and Requirements catalog. Individual responsibilities are in AFM 36-2664*, Personnel Assessment Program*. WAPS does not apply to the Air National Guard or Air Reserve Forces.

A3. Recommendations. Report unsatisfactory performance of individual course graduates to the 782 TRG/TGE, 917 Missile Road, Rm 1A300, Sheppard AFB TX 76311-2368 or E-mail 782csil@us.af.mil. Reference specific STS line item and/or paragraphs. For a quick response to problems, call the CSIL at DSN 736-2574 any time day or night.

SECTION B – COURSE OBJECTIVE LIST (COL)

B1. Measurement. Measure each learning objective as follows:

B1.1. Use of Progress Checks (PCs) & Rubrics. Indicates formal measurement of knowledge (K) and/or performance (P) elements.

B2. Standard. Standards for measurement indicate the course objectives and delineate the individual progress checklist and rubrics. The minimum standard is 70% on knowledge progress checks. Instructors assist students as the standard for performance progress checks and warranted during the progress check. Students may be required to repeat all or parts of the learning outcomes until the student attains satisfactory performance.

B3. Proficiency Level. Student must demonstrate mastery on each learning outcome/objective before progressing to the next learning requirement.

B4. Course Objective List. The COL lists the objectives in the sequence taught by Blocks of Instruction. Per AETCI 36 - 2651, *Basic Military and Technical Training*, Supervisors can request a detailed listing of the initial skills course objectives by written request through the requesting organizations MAJCOM to the 366th Training Squadron, 3E1X1 Training Manager.

SECTION C – SUPPORT MATERIAL

C1. Air Force Qualification Training Packages

C1.1. For a complete list of up-to-date AFQTPs applicable to the 3E1X1 AFSC, go to **myLearning** or **CE DASH** under the documents tab in the AFSC AFQTP folder.

C1.2. In addition to the paper-based AFQTPs there are web-based courses or assessments developed, for specific tasks available on **myLearning** under Civil Engineering in their specialty topic area.

C2. Career Development Course (CDC) Assessment for Civil Engineer CDC/DL Course.

C2.1. The RTM has developed CDC assessments for the 3E1X1 career field, and they are located on **myLearning** under Civil Engineering in CE CDC Assessments section.

C2.2. The sole purpose of CDC assessments is to provide the Unit Commander, UTM, and the supervisor, a predictive indicator of whether the trainee has studied sufficiently to successfully pass their CDC end of course (EOC) exam(s).

SECTION D – EDUCATION AND TRAINING COURSE INDEX

D1. Purpose. This CFETP section identifies training courses available for the HVAC/R specialty. Refer to the **(ETCA)** web site for information on the Air Force in-residence courses.

D2. Air Force In-Residence Courses/Mobile Training Team (MTT) Courses.

Course Number	Title	Developer
J8AQR3E131 01AB	HVAC/R Apprentice ITRO Qual 1	366 TRS
J3ABR3E131 03AD	HVAC/R Apprentice AF Unique	366 TRS
J3ACR3E171 00AB	HVAC/R Craftsman Course	366 TRS

D3. Air Force Career Development Academy (AFCDA).

Course Number	Title	Edit Code
CDC 3E151	HVAC/R Journeyman	01

D4. Exportable/Web-based Courses/Information.

Course Number	Title	Developer
Web based	130K BTU Heater Course	AFCEC/COF
Web based	3E1X1 Paper Based AFQTPs	AFCEC/COF
Web based	Arc Flash Safety Awareness AFQTP	AFCEC/COF
Web based	Civil Engineer 5-Level Core Concepts Course	AFCEC/COF
Web based	Civil Engineer 7-Level Core Concepts Course	AFCEC/COF
Web based	Confined Space Course	AFCEC/COF
Web based	HVAC/R Basic Electrical Diagnostics Course	AFCEC/COF
Web based	Industrial Water Treatment Orientation Program	AFCEC/COF
Web based	TRICON Refrigerated Container System (TRCS) Course	AFCEC/COF
Web based	WH-400 Water Heater Course	AFCEC/COF
WMGT 131	SMS Builder Course	AFIT
WENG 170	Cybersecurity for Control Systems	AFIT
WENG 200	Scoping and Estimating Course	AFIT
WENG 270	Advanced Control Systems Cybersecurity Course	AFIT
WMGT 301	Intro to Asset Management	AFIT
WMGT 322	Introduction to Project Management	AFIT
WENG 370	Control Systems Cybersecurity for CE Leaders	AFIT
WENG 400	Life-cycle Cost Estimating	AFIT
WMGT 436	Requirements and Optimization	AFIT
WMGT 437	Troop Construction Project Management	AFIT
WENG 460	Introduction to Mechanical Systems	AFIT
WENG 481	Contingency Facility Design	AFIT
WENV 531	Air Quality Management	AFIT
WENG 560	Fundamentals of HVAC Design and Analysis	AFIT
WENG 561	Application of HVAC Design and Analysis	AFIT
WENG 563	HVAC Control Systems	AFIT

D5. Courses/CDCs under Development/Revision

Course Number	Title	Date Due
J3AZR3E151 00RA	Advanced HVAC/R Cooling and Refrigeration Course	TBD
J3AZR3E151 00HA	Advanced HVAC/R Heating Course	TBD
J3AZR3E151 00BA	Advanced Air & Hydronic Balancing Course	TBD
J3ACR3E171 00AC	Advanced HVAC/R Craftsman Course	TBD
M-01	HVAC/R Journeyman	TBD

SECTION E – MAJCOM UNIQUE REQUIREMENTS

There are currently no MAJCOM unique requirements. This area is reserved.

SECTION F – HOME STATION TRAINING

F1. Purpose. This section aims to identify the tasks, training references, and sources available in support of contingency/wartime training. Civil Engineer forces will train to meet the full range of tasks expected in the contingency environment. Training ranges from knowledge-type training conducted in a classroom to task-oriented hands-on training conducted in the field.

F2. Foundational Training (FT). FT is knowledge-based and hands-on training conducted at the individual's home station for contingency operations. The CE Commander ensures training is provided, documented, and appoints subject matter experts to conduct training as required.

F3. Combat Skills Training (CST). CST is an integral part of the FT program. Lessons learned from past and current contingency operations have taught us the importance of maintaining a high level of combat readiness. Although the inclusion of combat skills-focused training into FT does not fully prepare CE personnel to work in a high-threat combat environment, the steps taken to enhance training will help elevate units to a readiness level capable of supporting safe and effective operations in low to medium-risk combat environments.

F4. Skills And Knowledge Training. Wartime or contingency environments often involve specialized and unique mission-essential equipment that civil engineers do not use daily. Mission essential contingency equipment and trainer expertise are unavailable at most CONUS installations due to the cost and complexity. Personnel must be hands-on certified, and the certification must be documented in their CFETP. AFI 10-210, *Prime Base Engineer Emergency Force (BEEF) Program*, Attachment 4, identifies the minimum number of trained personnel (positions) by specialty and the frequency requirements. Inadequate training on these key equipment items can negatively affect Air Force contingency operations.

F5. AF Expeditionary Training Requirement. The AF must train as it fights and continually assess expeditionary readiness training across the AF continuum of learning to produce Airmen ready to support all combatant commands. Expeditionary readiness training must be relevant, timely, synchronized, standardized, and integrated to ensure combatant commands provide a standard presentation of forces to support specified mission requirements while maximizing efficiency. Expeditionary readiness training divides training into three categories to ensure Airmen receive the right training at the right time: Basic Airman Readiness, Basic Deployment Readiness, and Advanced Deployment Readiness. For additional information, refer to AFI 10-405, *Ready Airmen Training.*

F6. Expeditionary Training References.

F6.1. AFI 10-209, *RED HORSE Program*, Chapter 3 and Attachments 4-9 identify RED HORSE training requirements.

F6.2. AFI 10-210, *Prime Base Engineer Emergency Force (BEEF) Program*, Chapter 4 and Attachments 2-8 identify the Prime BEEF training requirements.

F6.3. DAFMAN 36-2689, *Training Program*, and AFI *10-405*, *Ready Airmen Training*, identifies expeditionary readiness training requirements.

F6.4. Web-based Training products are available on **myLearning**. Airmen completing these courses receive credit for FT. Use group WBT products in a classroom setting to train as many personnel as possible. Document group training attendance on a sign-in roster IAW AFI 10-210.

BY ORDER OF THE SECRETARY OF THE AIR FORCE

OFFICIAL

TOM D. MILLER, Lieutenant General, USAF
DCS/Logistics, Installation and Mission Support

3 Attachments
1. Competency Based Learning Code Key
2. 3E1X1 Specialty Training Standard (STS)
3. 3E1X1 Supplemental Course Training Standards (CTSs)

Attachment 1
Competency Based Learning Code Key

A1. Competency Based Learning (CBL) Code Key

Code	Behavioral Statement Special Training Standard STS Coding System
K	**Subject Knowledge Training –** The verb selection identifies the individual's ability to identify facts, state principles, analyze or evaluate the subject.
P	**Performance Training –** Identifies that the individual has performed the task to the satisfaction of the course; however, the individual may not be capable of meeting the field requirements for speed and accuracy.
pk	**Performance Knowledge Training –** The verb selection identifies the individual's ability to relate simple facts, procedures, operating principles and operational theory for the task.
Blank	Specific tasks not identified with a symbol, or a behavioral code indicates that no training is provided in the course. Major commands and/or units may establish scale values and combat training as mission requirements dictate.

Symbol or Code	Explanations
5	This symbol in the Core task column indicates that it is a 5-level core task.
7	This symbol in the Core task column indicates that it is a 7-level core task.
9	This symbol in the Core task column indicates that it is a 9-level core task.
^	This symbol in the Core task column indicates that the Career Field Manager has mandated 3rd party task certification.
*	This symbol in the Deployment/SEI/TQT column indicates that the task is a deployment task.
+	This symbol in the Deployment/SEI/TQT column indicates that the task is a Special Experience Identifier.
~	This symbol in the Deployment/SEI/TQT column indicates that the task is CBRN Qualification Task.
P/pk	This mark in the course columns indicates training is taught, but not at the required level due to resource limitations. Hands-on training will be accomplished as soon as resources become available.
(I)	This mark in the course columns is used to show that training is multi-service.
♦	A diamond in the task column indicates it is a core task, however due to equipment or funding constraint at some units, the completion of the AFQTP and/or web-based training course is all that is required for upgrade. Hands-on certification must be accomplished at the first opportunity when equipment or funding is available.
TQT	TQT in the task column indicates the task is a CBRN Qualification Task. IAW AFI 10-2503, these tasks will also be accomplished in MOPP 4 and annotated on DAF Form 623A or DAF Form 797.
▲	A black triangle in the task column indicates an AFQTP is available for use.
△	A white triangle in the task column indicates an AFQTP is under development.
(AD)	AD in the task column indicates this core task is for Active Duty only; however, it is highly recommended for the ARC Component

Attachment 2
3E1X1 Specialty Training Standard (STS)

A2. Specialty Training Standard.

A2.1. Identification. In the training record User Profile section, the UTM will assign individuals to the correct workcenter, upon in processing into the unit.

A2.2. HVAC/R Specialty Tasks. The following are training requirements that the workcenter supervisor will use to track each duty position created for their workcenter.

A2.3. Wartime Training Posture. In response to a wartime posture, all tasks will be taught in the 3-level course at an accelerated rate in a streamlined environment. A "Wartime Training Posture" will be 6-days a week/10-hour day.

1. Tasks, Knowledge and Technical References	2. Tasks		3. Certification For OJT					4. Behavioral Codes Used To Indicate Training/Information Provided via DL and/or Course			
	A	B	A	B	C	D	E	A	B	C	D
	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
1.0. CIVIL ENGINEER (CE) CORE CONCEPTS COURSES TR: **myLearning**											
1.1. Complete CE 5-level Core Concepts Course	5								K		
1.2. Complete CE 7-level Core Concepts Course	7									K	
1.3. CE Core Concepts Overview								K			
1.4. Unpack, build, teardown, and repack Small Shelter System (SSS)								P			
1.5. Complete AFIT WMGT 131, SMS Builder Level 1											
1.6. Complete AFIT WENG 200 Scoping and Estimating Course **(AD)**	7										
2.0. SAFETY Description: To remain vigilant to one's safety, safety of others, and equipment. To recognize hazards associated with a task or environment, and to be cognizant of the inherent dangers of bodily harm, injury, death, or damage to equipment. **Supporting Competencies:** Self Awareness, Accountability, Verbal Communication, and Self-Control											
2.1. SAFETY BASIC **Required Behaviors:** 1) Adheres to Personal Protective Equipment (PPE) requirements and shop safety practices 2) Attends work center safety briefings and takes appropriate actions 3) Identifies potential hazards in work place environment **Method of Measurement:** Depth of Knowledge: Established practice with some workplace elements)											

Attachment 2
3E1X1 Specialty Training Standard (STS)

1. Tasks, Knowledge and Technical References	2. Tasks		3. Certification For OJT					4. Behavioral Codes Used To Indicate Training/Information Provided via DL and/or Course			
	A	B	A	B	C	D	E	A	B	C	D
	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
2.1.1. Describe how to remove a victim from an energized circuit								K(I)			
2.1.2. State the precautions that must be taken when installing refrigerant piping								K(I)			
2.1.3. Demonstrate the safety practices associated when performing Preventive Maintenance (PM) on HVAC equipment, components, and accessories	5^							P(I)			
2.1.4. Demonstrate the safety precautions associated with troubleshooting cooling systems	5^							P(I)			
2.1.5. State the safety precautions for handling and disposing of hazardous materials, including chemicals, contaminated oils, and compressed air and gasses								K(I)			
2.1.6. State the basic safety requirements for pressure-testing a system once it has been installed								K(I)			
2.1.7. Perform "Lock Out/Tag Out" procedures on cooling equipment	5^							P(I)			
2.1.8. Perform first aid procedures for electrical shock	5^							P(I)			
2.2. SAFETY INTERMEDIATE **Required Behaviors:** 1) Contributes to a culture of safety by leading by example 2) Corrects identified hazards **Method of Measurement:** Depth of Knowledge: Established practice with all workplace elements											
2.2.1. Obtain an OSHA 10 certificate											
2.2.2. Demonstrate the safety precautions used when working on hot-water systems	5^										
2.2.3. Identify proper safety procedures to follow on discovery of a gas leak											

Attachment 2
3E1X1 Specialty Training Standard (STS)

1. Tasks, Knowledge and Technical References	2. Tasks		3. Certification For OJT					4. Behavioral Codes Used To Indicate Training/Information Provided via DL and/or Course			
	A	B	A	B	C	D	E	A	B	C	D
	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
2.2.4. Identify proper safety procedures to follow on discovery of an oil leak											
2.3. SAFETY ADVANCE **Required Behaviors:** 1) Performs quality control checks to ensure compliance with shop safety 2) Seeks out ways to improve safety in the work place environment **Method of Measurement:** Depth of Knowledge: New practices of all workplace elements											
2.3.1. Analyze previous shop safety write-ups and verify corrections have been implemented and are effective											
2.3.2. Perform quality checks on job sites and in the work bay to ensure Airmen are complying with shop and unit safety policy											
2.3.3. Ensure sufficient PPE is available in the shop safety locker											
2.3.4. Ensure guards are properly secured on equipment and tools											
2.3.5. Identify relevant safety topics and brief shop personnel to aid in the development of a culture of safety	7^										
2.4. SAFETY EXPERT **Required Behaviors:** 1) Develops and implement safety practices for work center 2) Guides Airmen towards a culture of safety **Method of Measurement:** Depth of Knowledge: New practices/concepts and theories of all workplace elements; is a credible resource in this area											
2.4.1. Working with shop leadership, develop strategies to build a culture of safety											
3.0. AIR DISTRIBUTION & VENTILATION Description: Distribution and ventilation of air to and from a space through the use of fans, filters, ductwork and dampers **Supporting Competencies:** Troubleshooting, Design Sense, and Problem Solving											

Attachment 2
3E1X1 Specialty Training Standard (STS)

1. Tasks, Knowledge and Technical References	2. Tasks		3. Certification For OJT					4. Behavioral Codes Used To Indicate Training/Information Provided via DL and/or Course			
	A	B	A	B	C	D	E	A	B	C	D
	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
3.1. AIR DISTRIBUTION & VENTILATION BASIC **Required Behaviors:** 1) Performs PM on air distribution and ventilation systems (bearings, belts, filters, etc.) 2) Assists technicians with repairing and replacing components (motors, dampers, duct work, etc.) **Method of Measurement:** Consistency of Application: Sustained application of competency over time											
3.1.1. Describe and define the following: wet bulb temperature, dry bulb temperature, and dew point								K			
3.1.2. Describe the components of a Collective Protection (COLPRO) system								K			
3.1.3. Describe how to change the speed of an air distribution system supply fan								K			
3.1.4. Describe how to repair minor damage to Fiberglass duct								K			
3.1.5. Describe how to repair minor damage to Ductsox (Fabric Duct)								K			
3.1.6. Describe the air characteristics in a basic forced-air distribution system (static pressure, volume, density, etc.)								K			
3.1.7. Evaluate and replace Air Handling Unit (AHU) filters								P			
3.1.8. Evaluate, properly replace, adjust tension and align belts for an AHU								P			
3.1.9. Explain the differences between propeller, centrifugal fans and blowers								K			
3.1.10. Explain the importance for good indoor air quality, especially in a Sensitive Compartmented Information Facility (SCIF)								K			
3.1.11. Explain the installation of ductsox, flexible and fiber board duct								K			
3.1.12. Explain the need for a Dedicated Outside Air System (DOAS)								K			

Attachment 2
3E1X1 Specialty Training Standard (STS)

1. Tasks, Knowledge and Technical References	2. Tasks		3. Certification For OJT					4. Behavioral Codes Used To Indicate Training/Information Provided via DL and/or Course			
	A	B	A	B	C	D	E	A	B	C	D
	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
3.1.13. Explain the principles involved in balancing of an air distribution system								K			
3.1.14. Install takeoffs and attach flexible duct								P			
3.1.15. Recognize the kinds of air filters used with HVAC systems and explain why each is used								K			
3.1.16. Evaluate and lubricate a belt driven blower assembly bearing								P			
3.2. AIR DISTRIBUTION & VENTILATION INTERMEDIATE **Required Behaviors:** 1) Identifies and troubleshoot faulty air distribution and ventilation system components 2) Repairs and replace components to air distribution and ventilation systems 3) Installs air distribution system based on design criteria; use tools for air balancing such as manometers, flow hood, etc. **Method of Measurement:** Consistency of Application: Sustained application of competency over time in a variety of situations											
3.2.1. Perform preventative maintenance on an Air Handling Unit (AHU) using an AF PMTL	5^								pk		
3.2.2. Replace and adjust a sheave / pulley on a belt driven blower	5^								pk		
3.2.3. Explain different types of air compressor cooling (Inter Cooling & after Cooling)											
3.2.4. Describe how basic air compression technology works, including multi-stage compression											
3.2.5. Make air measurements using carbon dioxide (CO2) detector/sensor and explain the readings											
3.2.6. Describe procedures for installing components such as registers, diffusers, grilles, dampers, access doors, and zoning accessories											
3.3. AIR DISTRIBUTION & VENTILATION ADVANCE **Required Behaviors:** 1) Performs air balancing to meet design specifications											

Attachment 2
3E1X1 Specialty Training Standard (STS)

1. Tasks, Knowledge and Technical References	2. Tasks		3. Certification For OJT					4. Behavioral Codes Used To Indicate Training/Information Provided via DL and/or Course			
	A	B	A	B	C	D	E	A	B	C	D
	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
2) Assists with air system designs and plans 3) Upgrades air distribution and ventilation systems **Method of Measurement:** Consistency of Application: Sustained application of competency over time in complex situations											
3.3.1. Balance a selected air system:											
3.3.1.1. Measure fan and fan motor rpm											
3.3.1.2. Measure supply and return duct static pressures											
3.3.1.3. Use fan curve charts											
3.3.1.4. Adjust dampers in branch supply ducts and at air terminals and diffusers											
3.3.1.5. Identify system information for balancing report forms											
3.3.2. Calculate mixed air problems for infiltration and ventilation											
3.3.3. Define and use the Process Triangle on the psychrometric chart to calculate sensible heat										P	
3.3.4. Describe “R” values and application of various duct insulation materials											
3.3.5. Describe the procedure for finding cubic feet per minute using temperature rise											
3.3.6. Determine the number, location, and sizes of supply outlets and return inlets needed in a building											
3.3.7. Identify the various types of duct systems and explain why and where each type is used											
3.3.8. Identify & interpret a building indoor air quality inspection/evaluation										P	
3.3.9. List the occupant's symptoms of poor indoor air quality											

Attachment 2
3E1X1 Specialty Training Standard (STS)

1. Tasks, Knowledge and Technical References	2. Tasks		3. Certification For OJT					4. Behavioral Codes Used To Indicate Training/Information Provided via DL and/or Course			
	A	B	A	B	C	D	E	A	B	C	D
	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
3.3.10. Select air handling equipment for large deployable shelters used in an extremely hot/desert environment (K-SPAN, LAMS, B-huts, etc.)											
3.3.11. Determine duct size, using duct sizing formulas for ductwork to be used in a temporary shelter (K-SPAN, LAMS, B-Hut, etc.)										pk	
3.3.12. Describe the steps to install bare base equipment and ductwork in a LAMS per engineer drawings											
3.4. AIR DISTRIBUTION & VENTILATION EXPERT **Required Behaviors:** 1) Designs Air Distribution and Ventilation System for facilities/shelters 2) Guide technicians through detailed Air Distribution and Ventilation tasks 3) Coordinate with and advise engineers to finalize facility plans/blueprints **Method of Measurement:** Consistency of Application: Able to innovate and formulate strategies; able to model/guide/teach others the competency of how to apply the competency											
3.4.1. Explain the effect of fittings and transitions on duct system design											
3.4.2. Explain Fan Laws											
4.0. HYDRONICS Description: The use of water as a medium through piping and terminal units to transfer heat in cooling or heating systems. **Supporting Competencies:** Problem solving, Troubleshooting, and Design Sense											
4.1. HYDRONICS BASIC **Required Behaviors:** 1) Uses proper tools and equipment associated with system maintenance operations 2) Assists technicians with preventative maintenance (chemical treatment, cleaning strainers, etc.) to extend equipment life span and restore efficiency 3) Assists technicians with repairing and replacing components (pumps, valves, etc.) **Method of Measurement**: Consistency of Application: Sustained application of competency over time											
4.1.1. Dissemble, clean and reassemble a strainer								P			

Attachment 2
3E1X1 Specialty Training Standard (STS)

1. Tasks, Knowledge and Technical References	2. Tasks		3. Certification For OJT					4. Behavioral Codes Used To Indicate Training/Information Provided via DL and/or Course			
	A	B	A	B	C	D	E	A	B	C	D
	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
4.1.2. Explain the purpose of each component of hot-water heating, chilled-water cooling, and dual temperature water systems								K			
4.1.3. Explain the purpose of each component of steam-heating systems								K(I)			
4.1.4. Explain the terms and concepts used when working with chilled water-cooling hydronic systems								K			
4.1.5. Explain the terms and concepts used when working with hot water heating hydronic systems								K			
4.1.6. Explain the terms and concepts used when working with steam-heating systems								K(I)			
4.1.7. Fabricate a gasket								K			
4.1.8. Identify and explain various types of couplings, valves, strainers and fittings used with hydronic systems (including dielectric unions)								K(I)			
4.1.9. Identify common methods for measuring pipe								K			
4.1.10. Identify common types of pipes used in hydronic systems								K			
4.1.11. Describe components and functions of a centrifugal pump								K(I)			
4.1.12. Identify the common piping configurations used with steam-heating systems								K			
4.1.13. Identify the types of common piping configurations used with chilled-water systems including systems with cooling tower(s)								K			
4.1.14. Identify the types of common piping configurations used with hot-water systems								K			

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1. Tasks, Knowledge and Technical References	2. Tasks		3. Certification For OJT					4. Behavioral Codes Used To Indicate Training/Information Provided via DL and/or Course			
	A	B	A	B	C	D	E	A	B	C	D
	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
4.1.15. Identify types of gauges and explain how to obtain correct pressures								K			
4.1.16. Explain how to properly cut, ream, thread, and assemble black iron pipe								K			
4.2. HYDRONICS INTERMEDIATE **Required Behaviors:** 1) Troubleshoots and repair hydronic system to restore operation to design specifications 2) Repair, replace, fabricate, and install piping and components based on design criteria 3) Performs preventative maintenance to extend equipment life span and restore efficiency **Method of Measurement:** Consistency of Application: Sustained application of competency over time in a variety of situations											
4.2.1. Remove, install, and align centrifugal pumps and various types of pump couplings & seals	5^								pk		
4.2.2. Identify, explain, remove and install various types of gaskets and packing	5^								pk		
4.2.3. Turn on, operate, inspect and turn off a Hot water heating system											
4.2.4. Explain how to turn on, operate, inspect and turn off a Steam system											
4.2.5. Describe procedures for cleaning an open recirculating water system and related cooling towers											
4.2.6. Explain how to evaluate and install a steam trap											
4.2.7. Explain the reasons why water treatment programs are needed											
4.2.8. Explain cavitation and demonstrate how to remove air from a hydronic system	5^								pk		
4.2.9. Read the pressure across a water system circulating pump											
4.2.10. Explain how to Blowdown and clean a steam boiler											

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	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
4.2.11. Describe how to inspect and clean shell and tube condensers/evaporators and other water-type heat exchangers											
4.2.12. Perform glycol system tests (legionella in UK)											
4.2.13. Remove, fabricate and install black iron pipe and fittings	5^								pk		
4.3. HYDRONICS ADVANCE **Required Behaviors:** 1) Performs hydronic balancing to meet design specifications 2) Modifies hydronic systems to incorporate new equipment; determines requirements for equipment and component replacement **Method of Measurement:** Consistency of Application: Sustained application of competency over time in complex situations											
4.3.1. Balance water flow in a selected hydronic system:											
4.3.1.1. Adjust the devices used to measure and control water flow in hydronic systems											
4.3.1.2. Select a circulating pump based on required flow rates											
4.3.1.3. Demonstrate how to measure pump pressures and system flow rates in an operating system using as built drawings											
4.3.2. Explain how to determine the glycol to water ratio for freeze protection											
4.4. HYDRONICS EXPERT **Required Behaviors:** 1) Coordinates work with engineers to design hydronic systems and determine user's requirements 2) Guides technicians through detailed hydronic system tasks to include current and emerging technologies **Method of Measurement:** Consistency of Application: Able to innovate and formulate strategies; able to model/guide/teach others the competency of how to apply the competency											

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	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
4.4.1. Calculate flow rate requirements for new system											
4.4.2. Calculate the pipe size and length based on GPM											
4.4.3. Evaluate design diagrams for new system											
5.0. ELECTRICAL Description: Flow of electrons between positive & negative points through the use of conductors used to power HVAC/R & control systems for customer or process **Supporting Competencies:** Troubleshooting, Technology Savvy, and Problem Solving											
5.1. ELECTRICAL BASIC **Required Behaviors:** 1) Assists technician with electrical system tasks such as install, repair, and replacement 2) Performs minor preventive maintenance action (proper electrical connections, verified applied system voltage, amperage draws, capacitor reading, etc.) **Method of Measurement:** Consistency of Application: Sustained application of competency over time											
5.1.1. Define Alternating & Direct current								K(I)			
5.1.2. Define Amperage, Current, Resistance, Voltage, and Hertz								K(I)			
5.1.3. Define and identify conductors (wire color, types & applications)								K(I)			
5.1.4. Define Ohm's Law								K(I)			
5.1.5. Demonstrate how to isolate electrical problems to faulty power distribution, load, and control circuits								P(I)			
5.1.6. Describe and explain motor construction, speed, rotation for single phase and three phase motors								K(I)			
5.1.7. Describe and identify semi-conductors								K(I)			
5.1.8. Describe the differences between a "Pictorial", a "Ladder Diagram", and a "Schematic"								K(I)			

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	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
5.1.9. Explain the effects of voltage drop, amps, and resistance in a series, parallel and combination series-parallel circuit								K(I)			
5.1.10. Identify the types and describe the proper application and use of common switches used in HVACR								K(I)			
5.1.11. Identify the types and describe the proper rating and use of "Circuit Protectors"								K(I)			
5.1.12. Identify voltage between two points using schematic wiring diagrams								P(I)			
5.1.13. Install and verify proper operation of a control relay								P(I)			
5.1.14. Install and verify proper operation of a line starter								P(I)			
5.1.15. Install and verify proper operation of a transformer								P(I)			
5.1.16. Make voltage, current, and resistance measurements using electrical test equipment								P(I)			
5.1.17. Read and interpret common electrical symbols								K(I)			
5.1.18. Replace a 3-phase motor (wire for high and low volts)								P(I)			
5.1.19. Troubleshoot to identify the 3 common electrical faults (Shorts, Open, Overloaded)								P(I)			
5.1.20. Troubleshoot and replace start and run capacitors								P(I)			
5.2. ELECTRICAL INTERMEDIATE **Required Behaviors:** 1) Identifies and troubleshoot electrical system malfunctions 2) Replaces electrical system components and verify system operations 3) Installs electrical distribution and components based on interpretation of schematic & sequence of operation **Method of Measurement:** Consistency of Application: Sustained application of competency over time in a variety of situations											

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	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
5.2.1. Perform electrical tests and troubleshooting of compressors and motors (1 & 3 Phase)	5^								pk		
5.2.2. Describe the operation and characteristics of an Electronically Commutated Motor											
5.2.3. Determine the sequence of operation using schematic wiring diagrams											
5.2.4. Install a single and three phase motor and verify proper rotation; change rotation as required	5^								pk		
5.2.5. Install and verify proper operation contactors, capacitors, relays, thermostats, transformers, and defrost timers											
5.2.6. Install different types of motors (e.g. Shaded pole, split phase, PSC, Capacitor start capacitor run, and ECM) (3ph wire for Wye & Delta)	7^								pk		
5.2.7. Identify the electrical code requirements per the NEC or host nation for HVAC/R Systems											
5.2.8. Identify data plate information needed for troubleshooting & replacement											
5.2.9. Identify explosion proof regulations for any type of electrical equipment											
5.3. ELECTRICAL ADVANCE **Required Behaviors:** 1) Modifies HVAC/R electrical systems IAW regulations (NEC, NFPA, etc.) 2) integrates electrical system with industrial control system and verify system operation with ICS technicians 3) Performs advanced electrical troubleshooting (solid state components, PLC, DC voltage, frequency drive, etc.) **Method of Measurement:** Consistency of Application: Sustained application of competency over time in complex situations											
5.3.1. Isolate and correct malfunctions on a Chiller	7^										
5.3.2. Isolate and correct malfunctions on a Computer Room Air Conditioning (CRAC) unit											

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	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
5.3.3. Isolate and correct malfunctions on a domestic water heater											
5.3.4. Isolate and correct malfunctions on a Gas & Oil Furnace											
5.3.5. Isolate and correct malfunctions on a hot water and steam boiler	7^										
5.3.6. Isolate and correct malfunctions on an infrared heater											
5.3.7. Isolate and correct malfunctions on a Mini Split heat pump system											
5.3.8. Isolate and correct malfunctions on a Packaged AC unit	7^									P	
5.3.9. Isolate and correct malfunctions on a Variable Refrigerant Flow (VRF) unit											
5.3.10. Isolate and correct malfunctions on a Walk in Refrigerator/Freezer	7^										
5.3.11. Isolate and correct malfunctions on an ice machine											
5.3.12. Isolate and correct malfunctions on an industrial air compressor											
5.3.13. Isolate and correct malfunctions on a Make-up Air (MUA) unit											
5.4. ELECTRICAL EXPERT **Required Behaviors:** 1) Designs HVAC/R electrical systems in coordination with project manager and engineer 2) Provides mechanical and electrical expertise to project managers/engineers for modification and development of HVAC/R electrical systems **Method of Measurement:** Consistency of Application: Able to innovate and formulate strategies; able to model/guide/teach others the competency of how to apply the competency											
5.4.1. Determine power requirement and proper materials for a new HVAC equipment install per NEC or host nation											
5.4.2. Calculate wire sizing as it applies to voltage drop and length of wiring run per NEC or host nation											

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	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
6.0. CONTROL SYSTEMS (CS) Description: Control system to maintain, monitor, and regulate the energy consumption of a building by controlling the operation of energy consuming HVAC/R systems **Supporting Competencies:** Technology Savvy, Troubleshooting, and Problem Solving											
6.1. CONTROL SYSTEMS (CS) BASIC **Required Behaviors:** 1) Assists technician with industrial control tasks such as installing, repairing, and replacement 2) Performs preventative maintenance on industrial control systems **Method of Measurement:** Consistency of Application: Sustained application of competency over time											
6.1.1. Describe different types of thermostats and explain how they are used								K			
6.1.2. Explain the basic principles applicable to all control systems								K			
6.1.3. Explain the function of a controlled device in an HVAC system								K			
6.1.4. Explain the function of a controller in an HVAC system								K			
6.1.5. Explain the function of a sensor in an HVAC system								K			
6.1.6. Identify electrical and electronic components and circuits, recognize their diagram symbols, and explain their functions								K			
6.1.7. Explain the purpose of a Frequency Drive and describe how it operates								K			
6.1.8. Use different types of service instruments needed to troubleshoot control systems								P			
6.1.9. Explain Basic cybersecurity principles and terminology								K			
6.1.10. Identify roles and responsibilities pertaining to control systems cybersecurity								K			

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	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
6.2. CONTROL SYSTEMS (CS) INTERMEDIATE **Required Behavior:** 1) Initiates log in approval to controllers and make adjustments 2) Replaces control system components 3) identifies issues with controls sequence of operations **Method of Measurement:** Consistency of Application: Sustained application of competency over time in a variety of situations											
6.2.1. Install a programmable thermostat									pk		
6.2.2. Complete AFIT WENG 170 Cybersecurity for Control Systems Course WBT	5										
6.2.3. Demonstrate how to install typical sensors, actuators, power wiring, and communication wiring	5^										
6.2.4. Describe and test thermistor type & temperature sensors (PTC & NTC)	5^										
6.2.5. List the types of information available on a typical front-end computer screen for a Building Automation System											
6.2.6. Perform Preventative maintenance on a control system using an AF PMTL									pk		
6.2.7. Program and operate a basic direct digital controller	5^										
6.2.8. Explain how cybersecurity (confidentiality, integrity, availability) applies to control systems											
6.2.9. Follow administrative access best practices (Cybersecurity)											
6.2.9.1. Create a strong password that meets standards and explain how to store it properly (no sharing, no sticky notes, etc.)											
6.2.9.2. Describe how to treat control systems data and related reports as sensitive											

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	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
6.2.9.3. Explain policy on remote maintenance											
6.2.9.4. Describe importance of patching and system back-ups											
6.2.10. Follow physical access best practices (Cybersecurity)											
6.2.10.1. Describe importance of documenting who has access and control over control systems equipment (positive access control)											
6.2.10.2. Explain how to properly manage vendors (and other third parties) during on-site installation and maintenance											
6.3. CONTROL SYSTEMS (CS) ADVANCE **Required Behaviors:** 1) Programs and Repair controls components 2) Performs upgrades on systems 3) Analyzes control strategies to enable energy savings and proper sequence of operations 4) Troubleshoots programming and network of controls components 5) Coordinates with base or outside agencies to acquire updated software and cpyber security protocols **Method of Measurement:** Consistency of Application: Sustained application of competency over time in complex situations											
6.3.1. Complete AFIT WENG 563 HVAC Control Systems											
6.3.2. Evaluate and Install a Programmable Logic Controller (PLC)											
6.3.3. Interpret basic diagnostic, alarms and test data provided by a Building Automation System											
6.3.4. Adjust Building Automation System parameters and explain how to verify the change	7^										
6.3.5. Perform troubleshooting on Building Automation System and correct system malfunctions	7^										

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	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
6.3.6. Upload or reload a Building Automation System program & explain the data storage requirements											
6.3.7. Evaluate and program a Variable Speed Drive (VSD) & Variable Frequency Drive (VFD)	7^										
6.3.8. Identify the difference between system failures, tampering or cyber intrusion											
6.3.9. Identify resources and escalation criteria for control system incident response and recovery											
6.3.10. Reset User's Account											
6.3.11. Control systems design and cybersecurity (Cybersecurity)											
6.3.11.1. Explain control systems topology, components and terminology (IP vs Serial, server vs. endpoint, etc.)											
6.3.11.2. Describe how control systems design and architecture affect control systems cybersecurity											
6.3.11.3. Complete AFIT WENG 270 Advanced Control Systems Cybersecurity Course **(AD)**	7										
6.3.11.4. Complete AFIT WENG 370 Control Systems Cybersecurity for CE Leaders WBT	7										
6.3.12. Supply chain concerns and vulnerabilities (Cybersecurity)											
6.3.12.1. Explain how a supply chain applies to control systems acquisition											
6.3.12.2. Explain how nation-states and malicious actors can manipulate supply chain											
6.3.12.3. Identify restrictions surrounding certain systems/companies/countries of origin (blacklist, tested products list, etc.)											

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	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
6.3.13. Baseline cybersecurity policy/standards for design, acquisition, procurement, repair, construction (Cybersecurity)											
6.3.13.1. Explain cybersecurity considerations in requirements / development (baking in standards, costs, 1391s, etc.)											
6.3.13.2. Explain cybersecurity considerations in acquisition and procurement (RMF controls, proprietary systems, etc.)											
6.3.13.3. Identify policy on testing/patching control systems and coordinating with the cyber test range for control systems											
6.3.13.4. Explain category management for control systems (authorized systems list, supply chain management, etc.)											
6.3.13.5. Identify asset management best practices for control systems											
6.4. CONTROL SYSTEMS (CS) EXPERT **Required Behaviors:** 1) Designs control systems with proper sequence of operation to application 2) Configures entire network of controllers to facilities 3) Guides technicians through detailed controls tasks **Method of Measurement:** Consistency of Application: Able to innovate and formulate strategies; able to model/guide/teach others the competency of how to apply the competency											
6.4.1. Identify the typical steps required to install a Building Automation System and explain CE COINE											
6.4.2. Verify source, design, requirements, and components of control systems (HW/SW/firmware) (Cybersecurity)											

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6.4.2.1. Verify that control systems received/installed matches the requirements stipulated in contract											
6.4.2.2. Check components for suspicious devices/tampering											
6.4.2.3 Describe vendor acquisition/installation/maintenance directives outlined in the latest CE policy for control systems											
6.4.3. Inventory upkeep upon commissioning & repair (Cybersecurity)											
6.4.3.1. Identify which systems (BUILDER, ITIPS, eMASS, etc.) apply to control systems and which type of information is required											
6.4.3.3. Identify procedures for updating inventory systems upon commissioning and repair											
7.0. HEATING **Description:** The process of adding heat to a conditioned space through the utilization of radiation, convection, and conduction. **Supporting Competencies** Problem Solving, Troubleshooting, and Design Sense											
7.1. HEATING BASIC **Required Behaviors**: 1) Assists technician with heating system tasks such as installing, repairing, and replacement 2)Performs preventative maintenance on heating systems using prescribed checklists **Method of Measurement:** Consistency of Application: Sustained application of competency over time											
7.1.1. Describe, explain the function, evaluate, clean, and replace (when feasible) the following components:											
7.1.1.1. Burner primary safety control								K(I)			
7.1.1.2. Combination fan and limit switch								K(I)			
7.1.1.3. Door safety switch								K(I)			

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7.1.1.4. Flame sensor (Rods, CAD Cell, Infrared (IR), Ultraviolet (UV), IR/UV combo, electronic imaging								K(I)			
7.1.1.5. Flue baffles								K(I)			
7.1.1.6. Fuel oil pump								K(I)			
7.1.1.7. Gas piping drip-leg								K(I)			
7.1.1.8. Gas pressure regulating valves								K(I)			
7.1.1.9. Gas & oil valves								K(I)			
7.1.1.10. Heat exchanger								K(I)			
7.1.1.11. High voltage ignition transformer								K(I)			
7.1.1.12. Igniters (Electric Spark & Hot Surface Igniter (HSI)								K(I)			
7.1.1.13. Ignition module								K(I)			
7.1.1.14. In-shot burner								K(I)			
7.1.1.15. Oil pressure regulating valves **(TQT)**								K(I)			
7.1.1.16. Orifice								K(I)			
7.1.1.17. Pilot burner								K(I)			
7.1.1.18. Pressure gauges								K(I)			
7.1.1.19. Thermocouple and thermopile								K(I)			
7.1.1.20. Vent blower motor								K(I)			
7.1.1.21. Vent pressure switch								K(I)			
7.1.2. Define and differentiate between primary air and excess air								K(I)			
7.1.3. Define electric heat theory								K			
7.1.4. Demonstrate how to connect gas piping								P			
7.1.5. Safely perform operating procedures on a hot-water boiler								P(I)			
7.1.6. Safely perform operating procedures on gas fired systems, such as radiant heaters used in hangars								P			

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7.1.7. Describe and explain the function of upflow, downflow, and horizontal furnaces								K(I)			
7.1.8. Describe combustion theory, heating fuels and the byproducts of combustion								K(I)			
7.1.9. Describe how an electric furnace works								K(I)			
7.1.10. Explain Flue gas testing procedures for carbon monoxide								K(I)			
7.1.11. Explain the three methods by which heat is transferred and give an example of each								K			
7.1.12. Identify the tools and instruments used in troubleshooting oil heating systems								K(I)			
7.1.13. Identify the tools and instruments used when troubleshooting gas heating systems								K(I)			
7.1.14. Measure supply and return temperature and determine the temperature rise of a furnace								P			
7.1.15. Perform leak detection tests on gas systems **(TQT)**								P(I)			
7.1.16. Perform preventive maintenance procedures on a gas furnace, including filter replacement, cleaning of components, and temperature measurements								P(I)			
7.1.17. Perform operational test and explain the theory of operation of a 130K BTU Heater TO: 35E7-3-4-1 ♦	5^	*						P			
7.1.18. Explain the M-80 Water Heater theory of operation TO: 35E7-4-27-1 ♦	5^	*						K			
7.1.19. Perform operational test and explain the theory of operation of a WH-400 Water Heater TO: 40H3-7-1; TM 10-4520-266-13&P ♦	5^	*						P			

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7.1.20. Install and perform preventative maintenance on the 130K BTU Heater TO: 35E7-3-4-1 ♦	5^	*						P			
7.1.21. Explain the installation and preventative maintenance procedures for the M-80 Water Heater TO: 35E7-4-27-1 ♦	5^	*						K			
7.1.22. Install and perform preventative maintenance on the WH-400 Water Heater TO: 40H3-7-1; TM 10-4520-266-13&P ♦	5^	*						P			
7.2. HEATING INTERMEDIATE **Required Behaviors:** 1) Troubleshoots and repair heating equipment issues 2) Maintains equipment and operation logs 3) Installs, inspect, and replace heating equipment components 4) Performs seasonal overhaul on heating equipment **Method of Measurement:** Consistency of Application: (Sustained application of competency over time in a variety of situations											
7.2.1. Describe alternative technologies for heating, including:											
7.2.1.1. Geothermal											
7.2.1.2. Water-Source Heat Pumps											
7.2.2. Clean, start up, and shut down a hot water Boiler	5^								pk		
7.2.3. Demonstrate how to properly use a combustion analyzer											
7.2.4. Describe and state the causes of a lifting flame											
7.2.5. Describe and state the causes of burner "Flashback"											
7.2.6. Describe the fuel pressures in natural gas and liquefied petroleum (LP) gas piping											
7.2.7. Describe the oil pressures (and vacuum levels) on an operating oil-fired heating system											

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	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
7.2.8. Describe the procedure to perform a smoke test on an oil furnace											
7.2.9. Describe types and how a sequencer controls heating elements and blower operation											
7.2.10. Evaluate and replace a limit switch	5^										
7.2.11. Describe theory of operation, evaluation, installation, and set a pressure differential switch											
7.2.12. Explain complete and incomplete combustion											
7.2.13. Explain the various draft control devices used with natural-draft furnaces											
7.2.14. Test and adjust a barometric draft control											
7.2.15. Adjust various types of igniters (e.g. Pilot Burner & Electric Spark)	5^								pk		
7.2.16. Isolate and correct malfunctions in a gas and oil heating appliance (Control circuits, Combustion system, Safety controls, Air system, Condensing Boiler)								P			
7.2.17. Isolate and correct malfunctions on a 130K BTU Heater ♦	5^	*									
7.2.18. Isolate and correct malfunctions on a M-80 Water Heater ♦	5^	*									
7.2.19. Isolate and correct malfunctions on a WH-400 Water Heater ♦	5^	*									
7.3. HEATING ADVANCE **Required Behaviors:** 1) Performs full install of heating equipment and calculate heating requirements 2) Collaborates with outside agencies (e.g. manufacturers, contractors, inspectors, etc.) 3) Maintains PM program and schedule PM per manufacturers recommendation **Method of Measurement:** Consistency of Application: Sustained application of competency over time in complex situations											

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	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
7.3.1. Describe how to prepare a boiler for wet & dry storage											
7.3.2. Describe maintenance procedures for central heating plant											
7.3.3. Describe the content of flue gas and explain how it is vented											
7.3.4. Describe the techniques for venting different types of furnaces											
7.3.5. State the factors that must be considered when installing a furnace											
7.3.6. Describe the application and perform the installation of turning vanes											
7.3.7. Explain AF and local requirements for boiler inspections											
7.4. HEATING EXPERT **Required Behaviors:** 1) Coordinates work with engineers to design heating systems and determine user's requirements 2) Guides personnel on heating equipment and develops training plans **Method of Measurement:** Consistency of Application: Able to innovate and formulate strategies; able to model/guide/teach others the competency of how to apply the competency											
7.4.1. Calculate the correct size and type of furnace vent connector and metal vent using manufacturer's instructions, National Fuel Gas Code, American Gas Association specifications, or host nation											
7.4.2. Describe the procedure to de-rate a gas furnace at altitudes of 2,000 feet and above											
7.4.3. Explain the function of energy conservation devices and how they reduce energy cost											
7.4.4. Identify and describe the steps in the system design process											
8.0. COOLING & REFRIGERATION Description: The process of removing heat from the conditioned space through the utilization of various types of environmental control units											

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	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
Supporting Competencies: Problem Solving, Troubleshooting, and Technology Savvy											
8.1 COOLING & REFRIGERATION BASIC **Required Behaviors:** 1) Assists technician with cooling system tasks such as install, repair, and replacement 2) Performs minor preventive MX action (e.g. replace belts, air filter, insulation, etc.) **Method of Measurement:** Consistency of Application: Sustained application of competency over time											
8.1.1.1. Accumulator								K(I)			
8.1.1.2. Compressors (Centrifugal, reciprocating, scroll, rotary, and screw)								K(I)			
8.1.1.3. Condensers (air & water cooled)								K(I)			
8.1.1.4. Economizer								K(I)			
8.1.1.5. Evaporator pressure regulator (EPR)								K(I)			
8.1.1.6. Evaporator								K(I)			
8.1.1.7. Pressure controls								K(I)			
8.1.1.8. Suction & Dirge line								K(I)			
8.1.1.9. Liquid & suction line filter								K(I)			
8.1.1.10. Solenoid valves								K(I)			
8.1.1.11. Metering devices (capillary tube, thermostatic expansion valve, automatic expansion valve, electronic expansion valve)								K(I)			
8.1.1.12. Oil separator								K(I)			
8.1.1.13. Regulators								K(I)			
8.1.1.14. Receivers								K(I)			
8.1.1.15. Refrigerant distributors and feeder tubes								K(I)			
8.1.1.16. Sight glass								K(I)			
8.1.1.17. Vibration eliminator								K(I)			
8.1.2. Soldering, Brazing and Cutting											
8.1.2.1. Identify types of solder and alloys								K			

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8.1.2.2. Choose proper flux for each alloy								P			
8.1.2.3. Solder & braze tubing and fittings								P			
8.1.2.4. Use nitrogen or carbon dioxide when brazing								P			
8.1.2.5. Cut metal plate								P			
8.1.2.6. Set-up/use torch and equipment using various tips								P			
8.1.2.7. Indicate melting temperatures								K			
8..1.3. Describe alternative technologies for cooling, including:											
8.1.3.1. Ductless system (DX/hydronic)								K			
8.1.3.2. Computer room air conditioner (CRAC)								K			
8.1.3.3. Multi-zone								K			
8.1.4. Charge blended refrigerant into a system and explain why each piece of equipment is used								P(I)			
8.1.5. Convert temperature values between Celsius and Fahrenheit								P			
8.1.6. Correctly cut, bend and join copper tubing using compression, flare and swag fittings								P			
8.1.7. Correctly use the tools and instruments required for servicing cooling & refrigeration units								P(I)			
8.1.8. Define enthalpy and entropy								K(I)			
8.1.9. Define refrigerant recycling and reclaim								K			
8.1.10. Define vacuum and vacuum levels as required in the HVAC/R industry								K			
8.1.11. Define vaporization of a liquid, and its effect on heat								K(I)			

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	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
8.1.12. Demonstrate how to use a valve Core removal tool								P(I)			
8.1.13. Perform refrigerant recovery and explain the three basic recovery methods: liquid, vapor and push-pull								P(I)			
8.1.14. Describe a check valve used in a refrigeration unit, its function and operation								K			
8.1.15. Describe a chilled water system and its operation								K			
8.1.16. Describe a heat pump's design, configuration for both the heating and cooling cycle								K			
8.1.17. Describe a systematic approach for troubleshooting cooling systems and components								K			
8.1.18. Describe and define the following: BTU, latent heat, sensible heat, total heat								K(I)			
8.1.19. Describe and define the following: subcooled liquid, superheated vapor								K(I)			
8.1.20. Describe and define the following: wet bulb depression, wet bulb temperature, dry bulb temperature, and dew point								K			
8.1.21. Describe change of state between liquids, vapor, and solids								K(I)			
8.1.22. Describe condensation of a vapor, and its effect on heat								K(I)			
8.1.23. Describe crankcase heating methods and how they operate								K(I)			
8.1.24. Describe dry type evaporators and their operation								K(I)			
8.1.25. Describe how azeotropes and near-azeotropes differ from each other and from so-called pure refrigerants								K			

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	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
8.1.26. Demonstrate manual and automatic pump-down								P(I)			
8.1.27. Describe the color-coding scheme used to identify refrigerant cylinders								K			
8.1.28. Describe the following oils and their applications; mineral, alkyl, benzene, glycols, and esters								K			
8.1.29. Describe the principles of dehumidification and explain how humidity affects human comfort and facility life cycle								K			
8.1.30. Describe the state of refrigerant, and explain what occurs in each major component during normal operation								K(I)			
8.1.31. Describe the thermodynamics of refrigerants								K(I)			
8.1.32. Describe how to use the Unified Facilities Code (UFC) and International Mechanical Code (IMC) and explain how they are used on the job								K			
8.1.33. Explain Heat transfer (Convection, Conduction, and Radiation)								K(I)			
8.1.34. Explain how heat transfer occurs in cooling systems (Direct & Indirect systems)								K			
8.1.35. Explain the differences in refrigerants and applications in low-, medium-, and high-temperature refrigeration systems								K			
8.1.36. Explain the operating sequence of a refrigeration system								K			
8.1.37. Explain the thermodynamics of air and water vapor								K			

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	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
8.1.38. Identify the tools and materials necessary for performing preventive maintenance on a packaged AC unit								K			
8.1.39. Identify the types of micron gauges and demonstrate how they should be connected to measure evacuation levels								P			
8.1.40. Identify types of copper pipe used in refrigeration applications								K			
8.1.41. Interpret a P-T chart for an azeotrope refrigerant								P			
8.1.42. Obtain an EPA Universal Certification								P(I)			
8.1.43. Obtain a NATE RTW Certificate								P			
8.1.44. Obtain system pressures using Manifold, Bourdon and Electronic Gauges								P(I)			
8.1.45. Describe a Headmaster on a CRAC Unit and its operation								K			
8.1.46. State the reason why capillary tube systems require a critical charge								K(I)			
8.1.47. Use the common types of leak detectors (electronic, ultrasonic, liquid (bubble), and ultraviolet/fluorescent) and explain the advantages and disadvantages associated with each type when pinpointing a leak								P(I)			
8.1.48. Install and perform operational test and explain the theory of operation on Field Deployable Environmental Control Unit (FDECU -5 & -9) TO: 35E9-314-1 ♦	5^	*						P			
8.1.49. Install and Perform operational test and explain the theory of operation on Advanced Design Refrigeration (ADR 300) TO: 40R7-6-1 ♦	5^	*						P			

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	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
8.1.50. Perform operational test and explain the theory of operation on TRICON Refrigerated Containerized System (TRCS) TO: 40R-6-11, TM 10-8145-224-13-P-V-1-0-0 ♦	5^	*						P			
8.1.51. Perform preventative maintenance on ADR 300 ♦	5^	*						K			
8.1.52. Perform preventative maintenance on FDECU -5 & -9 ♦	5^	*						P			
8.1.53. Perform preventative maintenance on TRCS ♦	5^	*						P			
8.2 COOLING & REFRIGERATION INTERMEDIATE **Required Behaviors:** 1) Performs major preventive MX actions (e.g. chiller barrel cleaning, component replacement, oil filters, etc.) 2) Interprets wiring diagrams to locate and repair system faults 3) Performs seasonal overhaul on cooling equipment **Method of Measurement:** Consistency of Application: Sustained application of competency over time in a variety of situations											
8.2.1. Isolate and correct mechanical malfunctions in a cooling system:											
8.2.1.1. System-related compressor problems								pk			
8.2.1.2. Refrigerant overcharge and undercharge								pk			
8.2.1.3. Evaporator and condenser problems								pk			
8.2.1.4. Metering device problems								pk			
8.2.1.5. Non-condensibles and contamination								pk			
8.2.2. Isolate and correct mechanical malfunctions in a heat pump system using the correct tools and instruments:											
8.2.2.1. Reverse cycle heating function											
8.2.2.2. Defrost cycle											
8.2.2.3. Auxiliary electric heat											

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	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
8.2.3. Calculate perimeter, area, and volume											
8.2.4. Charge refrigerant into a system by the following methods: Weight, Subcooling, Superheat, and Charging pressure chart	5^							P	pk		
8.2.5. Demonstrate how to adjust a water regulating valve on water cooled systems											
8.2.6. Demonstrate how to properly maintain recovery equipment											
8.2.7. Demonstrate how to test oil for contamination											
8.2.8. Demonstrate how to use an oil pump											
8.2.9. Demonstrate proper soldering and brazing techniques	5^								pk		
8.2.10. Describe a Cascade and Compound system, its application and operation											
8.2.11. Describe Ambient temperature controls											
8.2.12. Describe cooling towers and their operating limitations											
8.2.13. Describe how to clean and sanitize ice machines											
8.2.14. Describe problems associated with expansion valves replacement											
8.2.15. Describe the basic cycles and operation of ice machines											
8.2.16. Describe the operation of the following defrost controls: mechanical, time/temperature, and solid state											
8.2.17. Describe the piping configuration for a multiple evaporator system											

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	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
8.2.18. Describe the procedures used to clean up a system after a compressor burnout											
8.2.19. Remove and install a compressor (5 tons or less system)	5^								pk		
8.2.20. Describe various types of geothermal water loops and their application								K			
8.2.21. Describe charging a system with multiple evaporators using the manufacturer's literature											
8.2.22. Evacuate a system using a vacuum pump and micron gauge, and explain deep vacuum and triple evacuation method	5^							P	pk		
8.2.23. Evaluate and explain how to replace capillary tubes											
8.2.24. Explain compressor lubrication methods											
8.2.25. Demonstrate proper care and storage of a gauge manifold assembly											
8.2.26. Identify the kinds of hangers and supports needed for refrigerant piping											
8.2.27. Install a condensing unit											
8.2.28. Install a drain pan heater											
8.2.29. Install a packaged unit for a refrigeration system											
8.2.30. Install a replaceable core liquid line drier											
8.2.31. Install and adjust a low-pressure switch used for temperature control											
8.2.32. Isolate and correct mechanical faults in refrigeration system											
8.2.33. Perform a compressor efficiency test											

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8.2.34. Perform nitrogen regulator installation and adjustment used for system leak-test								P			
8.2.35. Perform reversing valve replacement											
8.2.36. Set up an electric defrost schedule for a refrigeration appliance											
8.2.37. Use equipment manufacturer's troubleshooting aids to troubleshoot HVAC system components and accessories	5^								pk		
8.2.38. Use temperature and pressure measuring instruments to make readings at key points in the refrigeration cycle											
8.2.39. Use the manufacturer's guidance to perform preventive maintenance **(TQT)**	5^								pk		
8.2.40. Safely join tubing by using flare, swag and compression fittings and identify the proper torque setting	5^								pk		
8.2.41. Correctly cut and bend copper tubing	5^								pk		
8.2.42. Describe Refrigerant throttling valve											
8.2.43. Describe temperature glide											
8.2.44. Describe the operation and function of a flooded evaporator and its metering device											
8.2.45. Obtain a NATE Support Technician Certificate											
8.2.46. Describe the principle of operation of a thermostatic expansion valve used with and without check valves											
8.3.47. Install a mini split air conditioning unit											

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	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
8.3.48. Install a Commercial Packaged air conditioning unit	5^								pk		
8.2.49. Install an FDECU -3 to -9 ♦	5^	*							pk		
8.2.50. Install an ADR 300 ♦	5^	*							pk		
8.2.51. Install a TRCS ♦	5^	*							pk		
8.3. COOLING & REFRIGERATION ADVANCE **Required Behaviors:** 1) Calculates building heat load to properly size system 2) Installs new systems to meet user requirements 3) Analyzes facility/shelter design to identify and correct design flaws **Method of Measurement:** Consistency of Application: Sustained application of competency over time in complex situations											
8.3.1. Complete a load estimate to determine the heating and cooling load of a building											
8.3.2. Justify the best location for installing HVAC/R equipment										P	
8.3.3. Explain equipment cooling and the different temperatures and relative humidity's effect on Servers or sensitive test equipment											
8.3.4. Explain how to size, assemble, and install a 50-foot run of refrigerant lines											
8.3.5. Explain the comfort zone and the different temperatures and relative humidity's effect on human comfort											
8.3.6. Explain the procedures to retrofit a system from a CFC to an HFC, & an HCFC to an HFC											
8.3.7. Explain the range of capacities for a commercial air system											
8.3.8. Differentiate between various types of insulation for a structure or facility (KSPAN, B-hut, LAMS, etc.)											
8.3.9. Install a replaceable core suction line filter drier											

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8.3.10. Install a thermostatic expansion valve											
8.3.11. Install an electronic expansion valve											
8.3.12. Size and explain the installation of a buffer tank for a chiller install											
8.3.13. Install an externally equalized expansion valve, correctly placing the sensing bulb and equalizer tube											
8.3.14. Interpret the temperature and pressure relationships at key points in the refrigeration cycle (Pressure - Enthalpy Diagram)											
8.3.15. Isolate and correct mechanical malfunctions in a Cooling and a Refrigeration system (Coil leaks, oil issues, faulty boards, over/under charge, etc.)	7^										
8.3.16. Perform troubleshooting on contingency equipment using a TO's troubleshooting chart ♦	7^										
8.3.17. Perform a contingency load estimate to determine the heating and cooling load of temporary shelters (KSPAN, LAMS, B-Hut, etc.)	7^									P	
8.4. COOLING & REFRIGERATION EXPERT **Required Behaviors:** 1) Coordinates work with engineers to design cooling systems and determine user's requirements 2) Guides personnel on cooling equipment and develops training plans **Method of Measurement:** (Consistency of Application: Able to innovate and formulate strategies; able to model/guide/teach others the competency of how to apply the competency)											
8.4.1. Compare the applications, installation considerations, and advantages and disadvantages of each type of Cooling systems											

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8.4.2. Evaluate the steps in the system design process											
9.0. PROGRAMS Description: A set of measures or activities that follow a set of temporary or continuously changing guidelines **Supporting Competencies:** Verbal Communication, Organization, Accountability											
9.1. PROGRAMS BASIC **Required Behaviors:** 1) Follows regulations in all work related activities 2) Identifies deficiencies to program manager (if needed) **Method of Measurement:** Reach of influence: Individuals											
9.1.1. Describe how to prepare an unfired pressure vessel for inspection											
9.1.2. Identify the latest PMTLs & RS Means for your shop											
9.1.3. Properly check out and return shop equipment or tools (accountability)											
9.2. PROGRAMS INTERMEDIATE **Required Behaviors:** 1) Enforces program guidelines to meet compliance 2) Maintains and update program continuity and self-assessment 3) Mitigates and correct deficiencies, if needed 4) Briefs work center personnel on guidance to follow proper protocol **Method of Measurement:** Reach of influence: Unit/Groups											
9.2.1. Assist HVAC shop's PM program manager											
9.2.2. Assist shop APIMS program manager									pk		
9.2.3. Explain an HVAC shop's confined space master entry plan											
9.2.4. Properly fill out and obtain a confined space entry permit											

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	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
9.3. PROGRAMS ADVANCE **Required Behaviors:** 1) Provides leadership data to make risk-based decisions 2) Advocates for resources to improve work center **Method of Measurement:** (Reach of influence: Wing/Institutional											
9.3.1. Manage shop resources such as people, tools and equipment and advocate for additional resources											
9.4. PROGRAMS EXPERT **Required Behaviors:** 1) Proposes recommended changes to regulations 2) Assesses program to identify any impact on future changes 3) Validates program requirements 4) Coordinates with outside agencies to schedule SAV **Method of Measurement:** Reach of influence: MAJCOM/AF-level practices/within industry											
9.4.1. Identify and explain UFC 3-410-01 (HVAC Systems)											
9.4.2. Identify and explain UFC 4-010-06 (Cybersecurity of control systems)											
9.4.3. Identify and explain DAFGM 2021-32-01 (Civil Engineer Control Systems Cybersecurity) or current active DAFI/DAFGM											
9.4.4. Identify and explain AFMAN 32-1068 (Heating Systems and Unfired Pressure Vessels)											
9.4.5. Identify and explain AFMAN 32-7002 (Environmental Compliance and Pollution Prevention)											
9.4.6. Identify and explain AFMAN 32-1001 (Civil Engineer Operations)											
9.4.7. Identify and explain CE Playbooks applicable to HVAC shops											
9.4.8. Maintain an IT system to track Boiler Inspections											

Attachment 2
3E1X1 Specialty Training Standard (STS)

1. Tasks, Knowledge and Technical References	2. Tasks		3. Certification For OJT					4. Behavioral Codes Used To Indicate Training/Information Provided via DL and/or Course			
	A	B	A	B	C	D	E	A	B	C	D
	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
10.0. TRAINING Description: Actions to develop and increase knowledge, skills, abilities & other characteristics enabling successful execution of tasks to support mission requirements. **Supporting Competencies:** Creative Thinking, Verbal Communication, and Time Management											
10.1. TRAINING BASIC **Required Behaviors:** 1) Applies and utilizes training resources to increase development of HVAC/R skill sets 2) Provides and documents effective training **Method of Measurement:** Levels of Impact: Specific workplace tasks											
10.1.1. Explain the latest career path for an HVAC Airmen								K			
10.2. TRAINING INTERMEDIATE **Required Behaviors:** 1) Acquires training resources and time across HVAC/R spectrum to support mission effectiveness 2) Assesses and validates personnel are trained and ready to support mission requirements **Method of Measurement:** Levels of Impact: Specific workplace projects											
10.2.1. Describe the purpose of the 3E1X1 Wartime Training Standard (WTS) & Silver Flag Exercise											
10.2.2. Identify the current list of AFIT courses applicable to your Airmen											
10.2.3. Describe the purpose of CE QTPs and explain how and why they are created											
10.2.4. Research AFI 10-210 (Prime BEEF Program) for HVAC requirements	5^										
10.2.5. Research AFI 10-209 (RED HORSE Program) for HVAC requirements ♦	5^										
10.2.6. Explain how the 3E1X1 and 3E1X1WG CFETPs are created											

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3E1X1 Specialty Training Standard (STS)

1. Tasks, Knowledge and Technical References	2. Tasks		3. Certification For OJT					4. Behavioral Codes Used To Indicate Training/Information Provided via DL and/or Course			
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	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
10.3. TRAINING ADVANCE **Required Behaviors:** 1) Advocates for training resources; courses, technology, seminars, etc. to ensure Airmen have the correct knowledge, skills, and abilities to execute the mission 2. Ensures the right mix of competent Airmen to execute the mission **Method of Measurement:** Levels of Impact: Management decisions											
10.3.1. Research training events (e.g. courses and seminars) and develop a shop training budget											
10.3.2. Perform Quality Assurance on jobs/tasks to identify specific individual training needs											
10.3.3. Identify innovative training technologies to ignite training in a work center (i.e. VR, AR, trainers, online training courses, etc.)											
10.3.4. Maintain and schedule recurring training day for your shop (i.e. monthly, quarterly, etc.)											
10.3.5. Identify and explain the Depatment of the Air Force Enlisted Classification Directory (DAFECD) and War Mobilization Plan											
10.4. TRAINING EXPERT **Required Behaviors:** 1) Develops and implements strategy to deliver qualified Airmen to enhance productivity to execute the mission 2) Vectors personnel to broaden individual development and career field capabilities **Method of Measurement:** Levels of Impact: AF-level practices/within industry											
10.4.1. Develop and Implement Master Task List and Duty Task List											
11. WORK CENTER MANAGEMENT Description: Manages work center task requirements to include labor, logistics, and process management actions required to successfully complete objectives of leadership and organizations **Supporting Competencies:** Time Management, Verbal Communication, and Problem Solving											

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3E1X1 Specialty Training Standard (STS)

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	A	B	A	B	C	D	E	A	B	C	D
	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
11.1. WORK CENTER MANAGEMENT BASIC **Required Behaviors:** 1) Follows guidelines on policies implemented by the work center 2) Reports any discrepancies in the work center 3) Performs assigned tasks **Method of Measurement:** Reach of influence: Individuals											
11.1.1. Identify and balance HVAC Preventive Maintenance requirements (manpower, equipment, materials, etc.)	5^								pk		
11.1.2. Complete AFIT WMGT 301 Intro to Asset Management **(AD)**	5										
11.1.3. Complete NexGen IT Training ♦	7										
11.2. WORK CENTER MANAGEMENT INTERMEDIATE **Required Behaviors:** 1) Maintains established programs 2) Prioritizes, schedule and assign work 3) Follow-ups with POCs to clarify objectives or instructions **Method of Measurement:** Reach of influence: Unit/Groups											
11.2.1. Estimate time for a job (fabrications or repairs)											
11.2.2. Complete AFIT WENG 460 Introduction to Mechanical Systems											
11.2.3. Complete AFIT WMGT 322 Intro to Project Management	7										
11.3. WORK CENTER MANAGEMENT ADVANCE **Required Behaviors:** 1) Incorporates leadership's vision into the mission 2) Advocates to leadership on current and future resource requirements 3) Standardizes shop procedures **Method of Measurement:** Reach of influence: Wing/Institutional											
11.3.1. Review and explain inspections in BUILDER											
11.3.2. Describe SMS Principles											
11.3.3. Complete AFIT WMGT 436 Requirements and Optimization											

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3E1X1 Specialty Training Standard (STS)

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	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
11.3.4. Propose facility mechanical projects for BCAMP submission											
11.3.5. Communicate risk in mechanical infrastructure and quantify the effects of resource decisions											
11.3.6. Assess a facility 30/60/90% design for maintainability and technical acceptance, specifically new shop maintenance requirements											
11.3.7. Complete AFIT WENG 437: Troop Construction Project Management Course **(AD)**	9										
11.3.8. Complete AFIT WENG 481: Contingency Facility Design course											
11.4. WORK CENTER MANAGEMENT EXPERT **Required Behaviors:** 1) Develops best practices that enhances overall workplace efficiency 2) Continuously seeks resources and ways to improve processes required to maintain, advance, and train personnel **Method of Measurement:** Reach of influence: MAJCOM/AF-level practices/within industry											
11.4.1. Develop best management practices that enhance overall workplace efficiency										P	
11.4.2. Complete AFIT WMGT 422 Project Management											
11.4.3. Complete AFIT WENG 400 Life-Cycle Cost Estimating											
11.4.4. Complete AFIT WENG 560 Fundamentals of HVAC Design and Analysis											
11.4.5. Complete AFIT WENG 561 Application of HVAC Design and Analysis											
11.4.6. Complete AFIT WENV 531 Air Quality Management Course											
11.4.7. Explain AF's facility condition assessment strategy and validate existing mechanical system data											

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3E1X1 Specialty Training Standard (STS)

1. Tasks, Knowledge and Technical References	2. Tasks		3. Certification For OJT					4. Behavioral Codes Used To Indicate Training/Information Provided via DL and/or Course			
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	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
12. SUPERVISION Description: Coordinates, manages, and leads individuals and/or sections' collective efforts to successfully accomplish organizational goals **Supporting Competencies:** Mentoring, Listening, and Develops People											
12.1. SUPERVISION BASIC **Required Behaviors:** 1) Observes and evaluate individuals to improve performance 2) Enforces AF standards to maintain accountability 3) Provides clear, consistent, and constructive feedback **Method of Measurement:** Reach of influence: Individuals											
12.1.1. Communicate with stakeholders (i.e. customers, supervisor, NCOIC, facility managers and contractors)	7^										
12.1.2. Perform routine job site visits and provide timely constructive feedback to subordinates											
12.1.3. Describe 3E1X1's role in a Basic Recovery After Attack (BRAAT) environments											
12.1.4. Identify the Equipment UTCs for 3E1X1s and General Consolidated Tool Kits (CTKs)											
12.1.5. Routinely assess and rotate crews (DIN, PM, Install, etc.) to deliberately develop Airmen											
12.2. SUPERVISION INTERMEDIATE **Required Behaviors:** 1) Communicates importance of new policy and procedures to influence personnel 2) Aligns and communicates strategic vision 3) Provides mentorship to personnel to become your successor 4) Utilizes available resources to develop Airmen 5. Develops and leads team building or networking activities within the section **Method of Measurement:** Reach of influence: Unit/Groups											
12.2.1. Incorporate MWR shop events for team building and networking activities											
12.2.2. Establish a SMART shop recognition program											

Attachment 2
3E1X1 Specialty Training Standard (STS)

1. Tasks, Knowledge and Technical References	2. Tasks		3. Certification For OJT					4. Behavioral Codes Used To Indicate Training/Information Provided via DL and/or Course			
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	Core Cert ^	Deployment * SEI +	Tng Start	Tng Comp	Trainee Initials	Trainer Initials	Certifier Initials	3 Lvl	5 Lvl	7 Lvl	9 Lvl
12.3. SUPERVISION ADVANCE **Required Behaviors**: 1) Communicates concerns and solutions to leadership that affects personnel 2) Develops strategies to enhance organizational climate **Method of Measurement:** Reach of influence: Wing/Institutional											
12.3.1. Develop and articulate a shop vision											
12.3.2. Analyze PM vs. CM including any predictive MX trends to make the best life-cycle decisions and develop a strategy to balance the shop or work center											
12.3.3. Evaluate HVAC/R requirements for a bare base environment and provide solutions to leadership (Equipment, tools, personnel, vehicles, etc.) (Contingency)											
12.4. SUPERVISION EXPERT **Required Behaviors:** 1) Communicates with Higher Headquarters (HHQ)leaders for growth and development opportunities 2) Aligns organization's goals with the goals of HHQ 3) Actively seeks to mentor personnel **Method of Measurement:** Reach of influence: MAJCOM/AF-level practice/within industry											
12.4.1. Complete AFIT WMGT 570 Civil Engineer Superintendent Course											
12.4.2. Access the Unit Manpower Personnel Record (UMPR) and compare to the Unit Manning Document (UMD)											
12.4.3. Utilize the budget cycle for unit (e.g. Form 9, GPC Unfunded List, Spend Plan)											

Attachment 3
3E1X1 Course Training Standards (STSs)

Supplemental Courses

ADVANCED HVAC/R BUILDING AUTOMATION SYSTEMS
J3AZR3E151 00CA (5 Days)

Behavior	CODE
1.0. Control Systems (CS) Intermediate	
1.1. Program and operate a basic direct digital controller	P
1.2. Explain how cybersecurity (confidentiality, integrity, availability) applies to control systems	K
2.0. Control Systems (CS) Advanced	
2.1. Interpret basic diagnostic, alarms and test data provided by a Building Automation System	P
2.2. Adjust Building Automation System parameters and explain how to verify the change	P
2.3. Troubleshoot Building Automation System and correct system malfunctions	P
2.4. Evaluate and Program a Variable Speed Drive (VSD) or Variable Frequency Drive (VFD)	P

Advanced HVAC/R Cooling & Refrigeration Course
J3AZR3E151 00RA Days (5-Days Proposed)

Behavior	CODE
1.0. Electrical Advanced	
1.1. Isolate and correct malfunctions on a chiller	P
1.2. Isolate and correct malfunctions on a Computer Room Air Conditioning unit	P
1.3. Isolate and correct malfunctions on a Variable Refrigerant Flow (VRF) unit	P
1.4. Isolate and correct malfunctions on a Walk in Refrigerator/Freezer	P
2.0. Cooling & Refrigeration Advanced	
2.1. Isolate and correct mechanical malfunctions in a Cooling and a Refrigeration system (e.g. Coil leaks, oil issues, faulty boards, and over/under charged)	P

Attachment 3
3E1X1 Course Training Standards (STSs)

Advanced Air & Hydronic Balancing Course
J3AZR3E151 00BA (5-Days Proposed)

Behavior	CODE
1.0. Air Distribution & Ventilation Advanced	
1.1. Balance a selected air system	
1.1.1. Measure fan and fan motor revolutions per minute (RPM)	P
1.1.2. Measure supply and return duct static pressures	P
1.1.3. Use fan curve charts	P
1.1.4. Adjust dampers in branch supply ducts and at air terminals and diffusers	P
1.1.5. Identify system information for balancing report forms	K
2.0. Hydronics Advanced	
2.1. Balance water flow in a selected hydronic system	
2.1.1. Measure and control water flow in hydronic systems using the correct equipment	P
2.1.2. After calculating required flow rates, select the correct circulating pump	P
2.1.3. Demonstrate how to measure pump pressures and system flow rates in an operating	P
3.0. Cooling & Refrigeration Advance	
3.1. Size and explain the installation of a buffer tank for a chiller install	P

Advanced HVAC/R Heating Course
J3AZR3E151 00HA Days (5-Days Proposed)

Behavior	CODE
1.0. Electrical Advanced	
1.1. Isolate and correct malfunctions on a hot water and steam boiler	P
1.2. Isolate and correct malfunctions on an infrared heater	P
2.0. Cooling & Refrigeration Advanced	
2.1. Isolate and correct malfunctions in a gas and oil-heating appliance (Control circuits, Combustion system, Safety controls, Air system, Condensing Boiler)	P